



IT3021 – Data Warehousing & Business Intelligence

BSc(Hons) in Information Technology

Specialized in Data Science

3rd Year

Faculty of Computing

SLIIT

2026 – Assignment 1

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IT Number : IT23725010

Task 01: Data Set Selection

The selected dataset is the Toy Store E-Commerce dataset from Maven Analytics. It represents a fully operational Online Transaction Processing (OLTP) system for a toy retail business that sells stuffed animal products online across multiple countries.

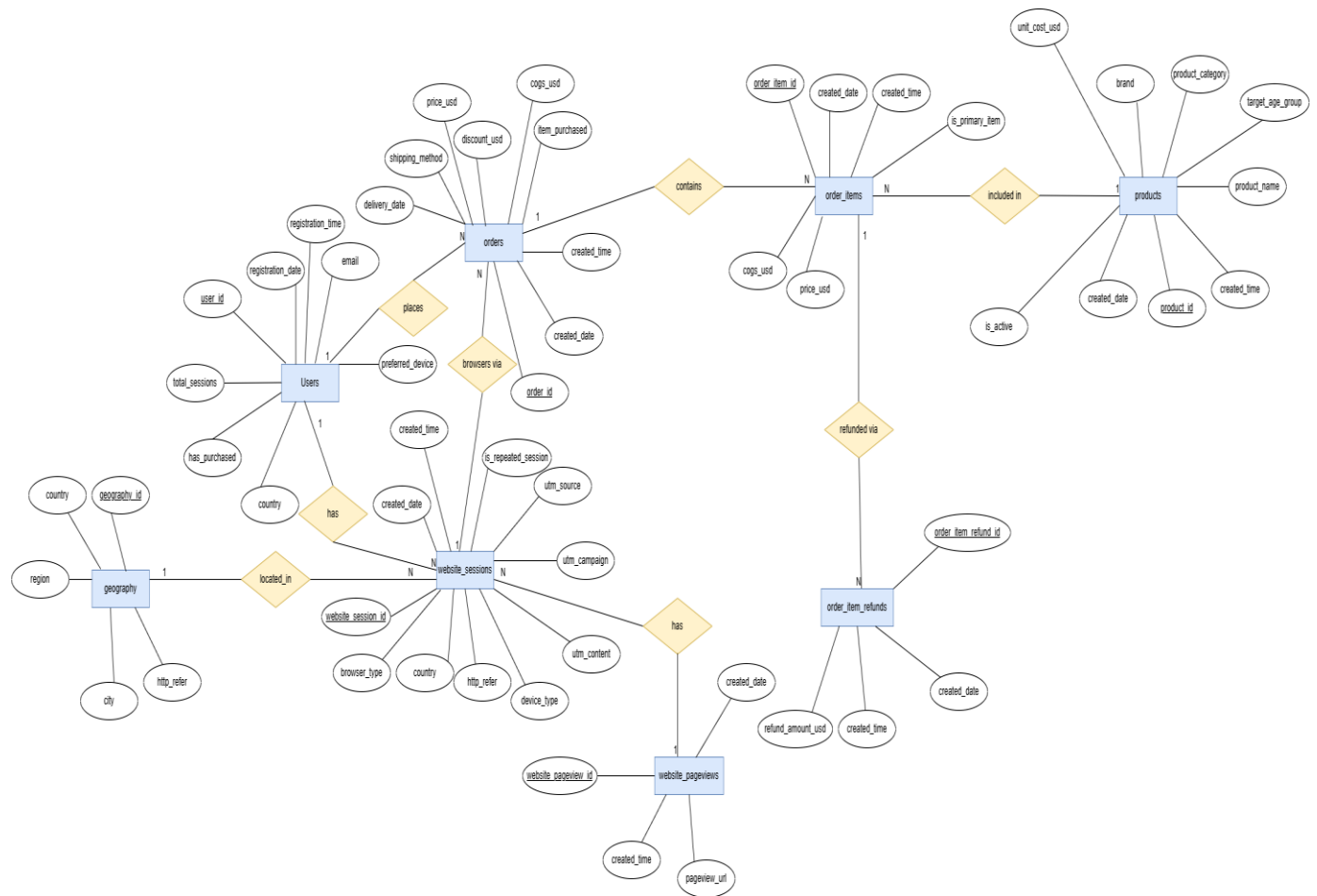
This is a genuine OLTP dataset — every table records raw, real-time business event data exactly as it would exist in a live operational system. The dataset does NOT contain any pre-built fact tables, dimension tables, or aggregated measures. It consists entirely of transactional and operational tables, making it an ideal source for designing a Data Warehouse from scratch. The data spans March 2012 to March 2015, providing exactly 3 years of transactional history, which far exceeds the 1-year minimum requirement. The dataset contains approximately 1.7 million total rows across 8 source tables: **orders**, **order_items**, **order_item_refunds**, **users**, **website_sessions**, **website_pageviews**, **products**, and **geography**.

The screenshot shows the Kaggle dataset page for 'Toy Store E-Commerce Database' by Siddharth0935. The page includes a sidebar with navigation options like Home, Competitions, Datasets, Models, Benchmarks, Game Arena, Code, Discussions, Learn, and More. The main content area displays the dataset's details, including a search bar, a 'Data Card' tab, and a 'Download' button. A table titled '_fuzzy_factory_data_dictionary.csv' (2.54 kB) is shown, detailing the structure of the dataset. The table has three columns: Table, Field, and Description. The 'Table' column lists 'users', 'website_sessions', 'orders', and 'orders'. The 'Field' column lists 'user_id', 'created_at', 'order_id', and 'website_session_id'. The 'Description' column provides details about each field, such as 'Identifier for the user. (PK)', 'Timestamp when the order was placed', and 'Unique identifier for each order (PK)'. A 'Data Explorer' sidebar on the right shows a list of files and a summary of the dataset.

Table	Field	Description
users	user_id	Identifier for the user. (PK)
website_sessions	created_at	Timestamp when the order was placed
orders	order_id	Unique identifier for each order (PK)
orders	website_session_id	Unique identifier for the website

Data set link: <https://www.kaggle.com/datasets/siddharth0935/toy-store-e-commerce-database>

ER Diagram

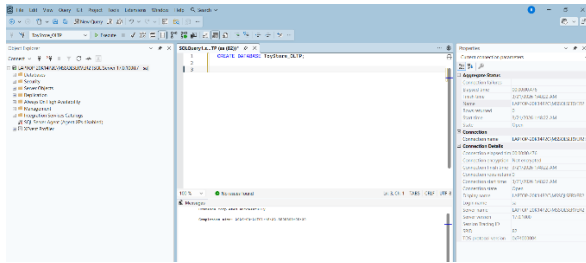


Task 02: Preparation of data sources

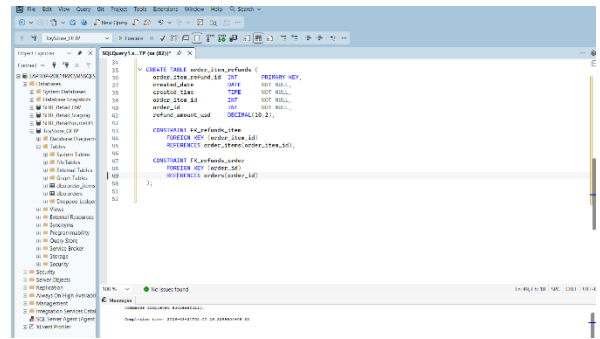
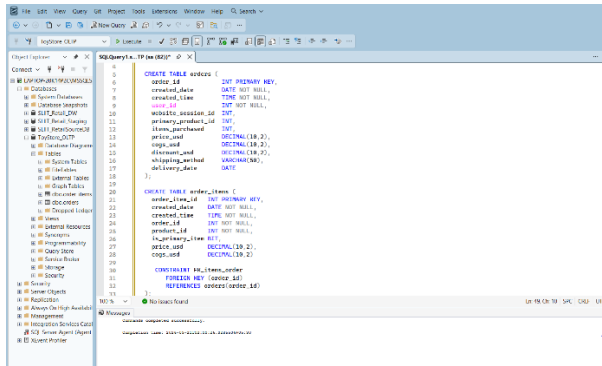
Source	Type	Files	SSIS Component	Business Justification
Source 1	SQL Server Database (ToyStore_OLTP)	orders order_items order_item_refunds website_sessions	OLE DB Source	Core transactional data lives in the operational database — standard e-commerce architecture
Source 2	CSV Flat Files	users.csv website_pageviews.csv	Flat File Source	CRM and web analytics data exported daily as CSV feeds from tracking and customer systems
Source 3	Excel Spreadsheets (.xlsx)	products.xlsx (enriched) geography.xlsx (enriched)	Excel Source	Product catalogue and geography master data maintained by the business team in Excel, enriched with hierarchy columns

2.1 SQL Server database setup

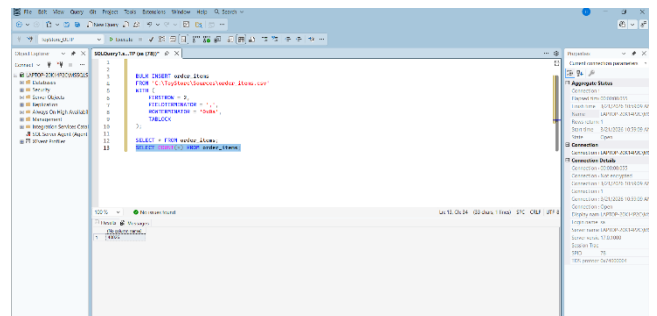
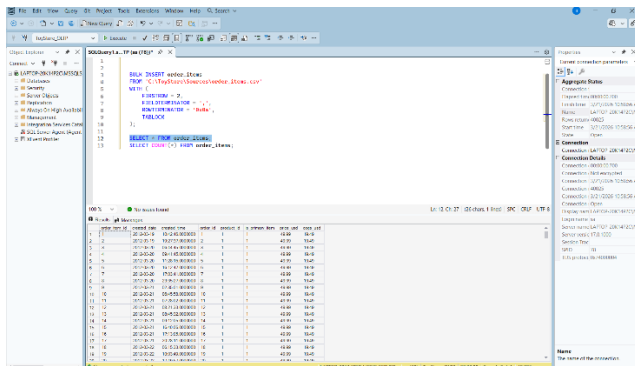
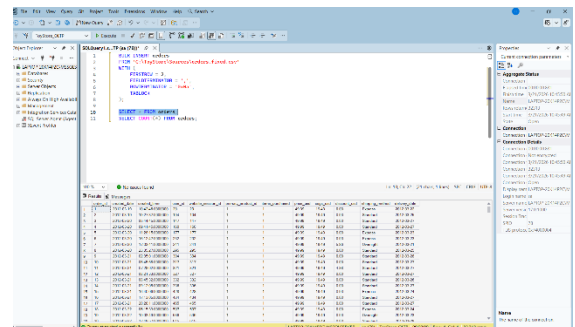
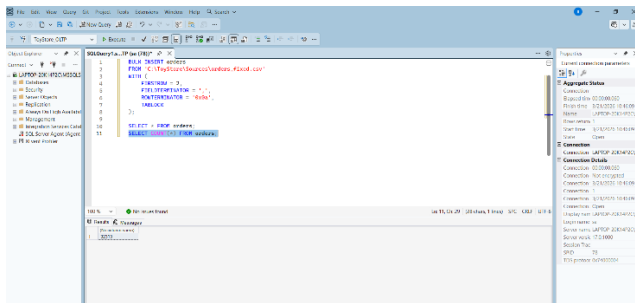
- Create ToyStore_OLTP database.

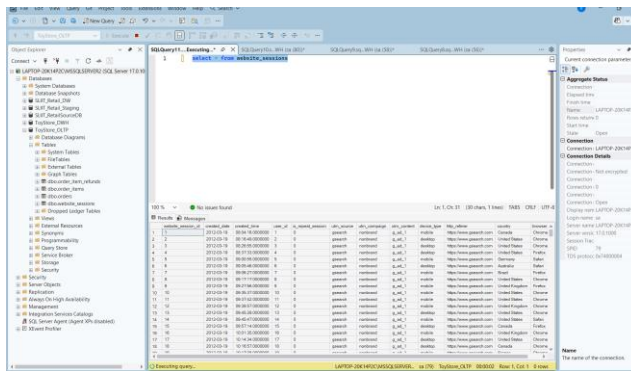
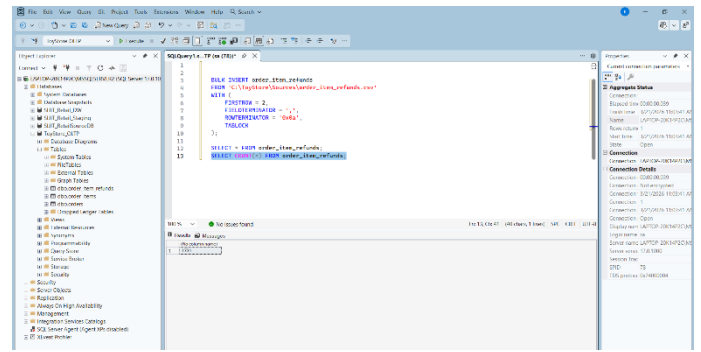
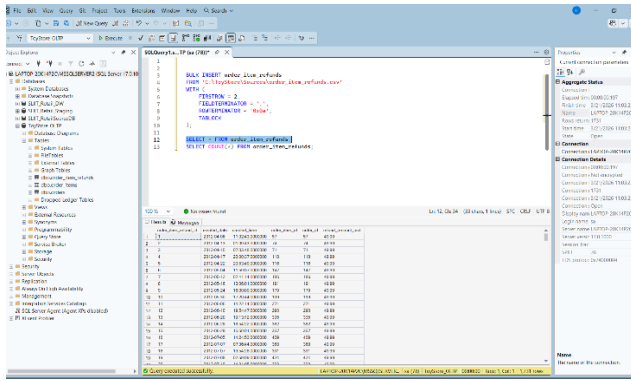


- After create orders, order_items, order_item_refunds table inside the database.



- After run and retrieve the tables with rows.





2.2 Excel files setup

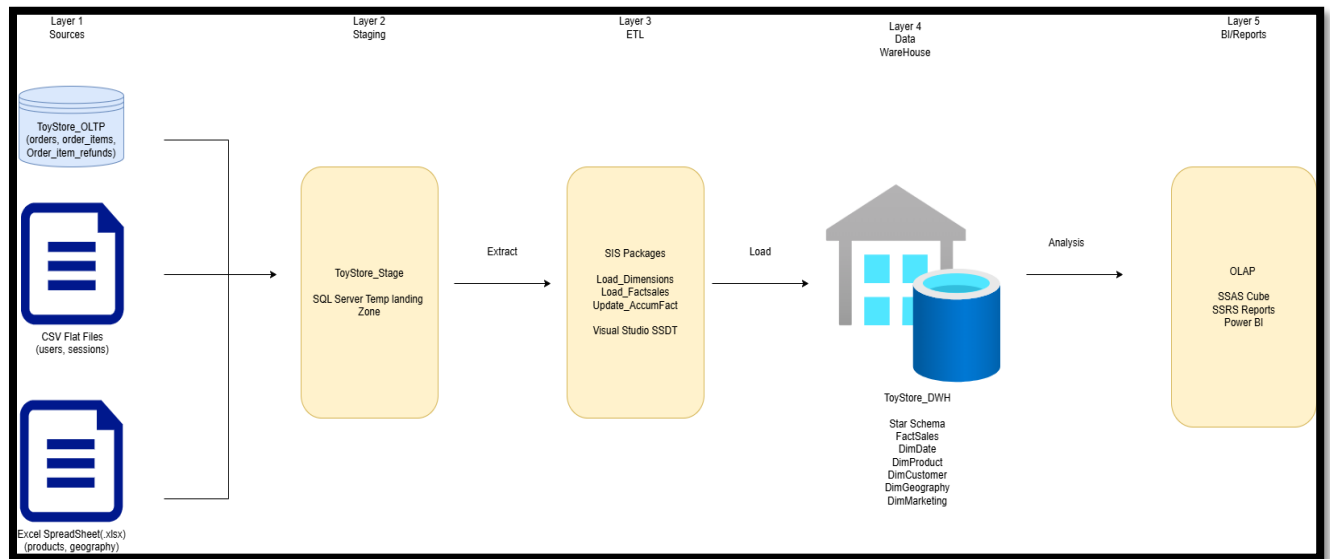
product_id	product_name	product_number	unit_cost_usd	target_age_group	is_active
1	The Original Mr. Fuzzy	Stuffed Animals	FuzzyWorld	19.49 4-8 years	1
2	The Forever Love Bear	Stuffed Animals	FuzzyWorld	24.99 4-8 years	1
3	The Birthday Sugar Panda	Stuffed Animals	SweetBears	17.99 0-3 years	1
4	The Hudson River Mini bear	Stuffed Animals	HudsonToys	11.99 0-3 years	1

Save As → Excel Workbook (.xlsx) → save as: C:\ToyStore\Sources\Excel\products.xlsx

File Home Insert Page Layout Formulas Data Review View Tell me what you want to do... Sign in Share																	
Clipboard		Font		Alignment		Number		Styles		Cells		Editing					
Paste		B I U		Merge & Center		\$ %		Conditional Formatting		Insert		Delete		Format		Autosum	
Format Painter										Normal		Bad		Good		Σ AutoSum	
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1 geography_id country region city http_referer continent																	
2 1 United States North America New York https://www.google.com North America																	
3 2 United States North America Los Angeles https://www.google.com North America																	
4 3 United States North America Chicago https://www.bing.com North America																	
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Save As: C:\ToyStore\Sources\Excel\geography.xlsx

Task 03: Solution Architecture



Layer 1- Data Sources:

- Three source types: SQL Server database, CSV flat files, and Excel spreadsheets
- SQL Server (ToyStore_OLTP) contains orders, order_items, and order_item_refunds
- CSV files contain users, website_sessions, and website_pageviews
- Excel files contain products and geography (enriched with hierarchy columns)

Layer 2- Staging Area:

- Temporary SQL Server database named ToyStore_Stage
- Raw data lands here before any transformation is applied
- Isolates extraction errors from the data warehouse
- Allows rollback if transformation fails

Layer 3- ETL(SSIS):

- Built using SQL Server Integration Services (SSIS) in Visual Studio
- Three packages: Load_Dimensions.dtsx, Load_FactSales.dtsx, Update_AccumFact.dtsx
- Handles transformations: SCD Type 2, Lookups, Derived Columns, Aggregations
- Loads dimensions first, then facts — order is enforced using Precedence Constraints

Layer 4- Data Warehouse:

- SQL Server database named ToyStore_DWH
- Star Schema design with one fact table (FactSales) and six dimension tables

- DimCustomer and DimProduct use SCD Type 2 for historical tracking
- Grain defined as one row per order line item

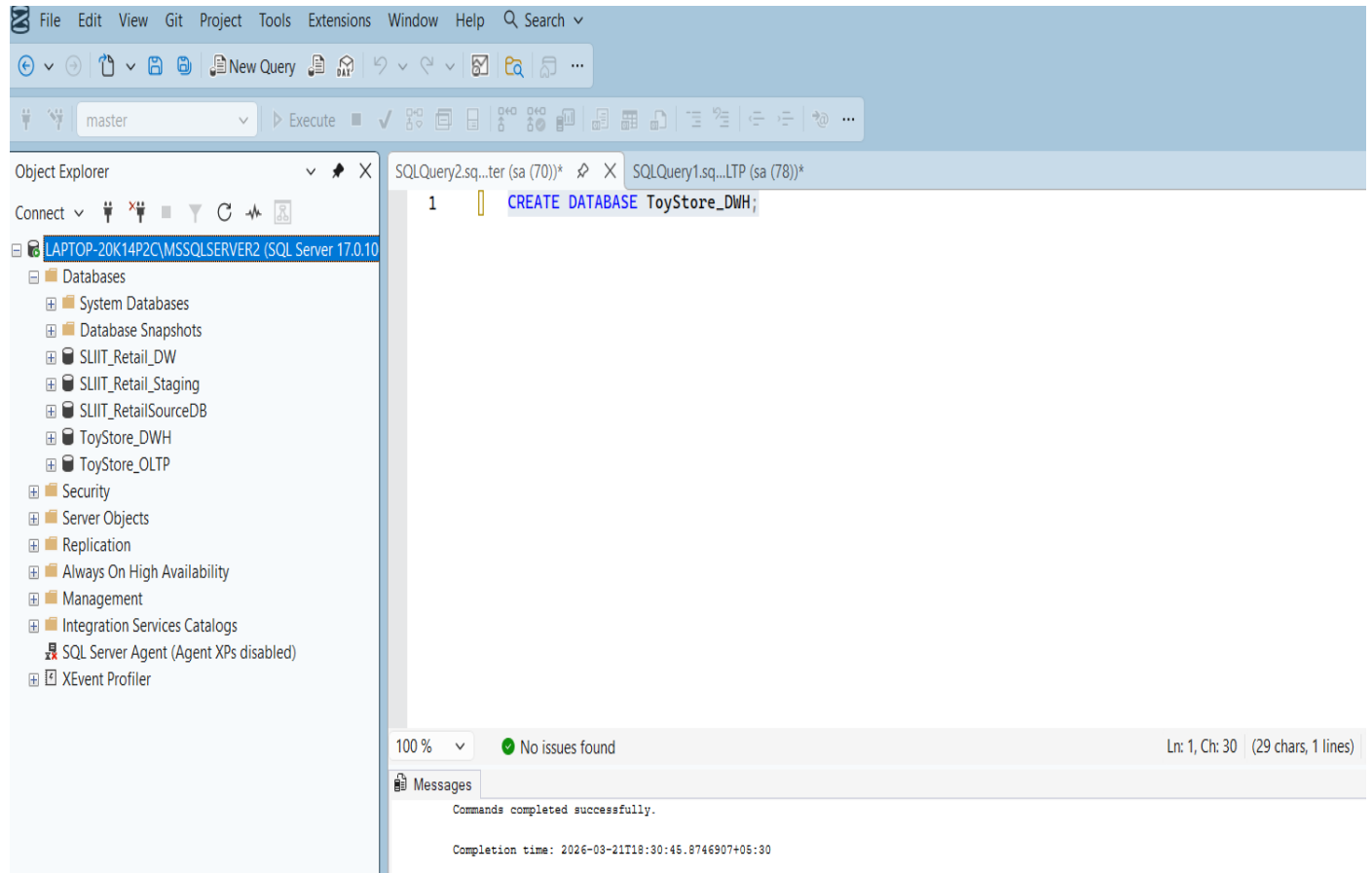
Layer 5- BI and Reporting:

- OLAP cube built using SQL Server Analysis Services (SSAS)
- Reports created using SSRS or Power BI

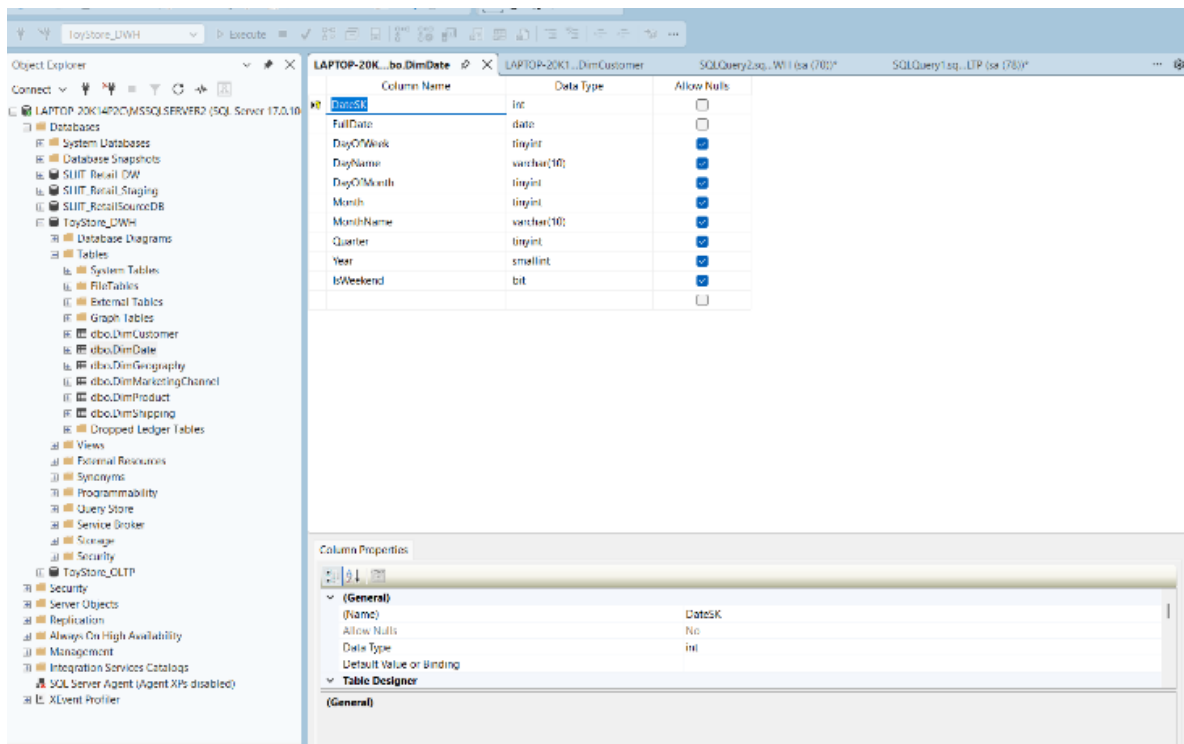
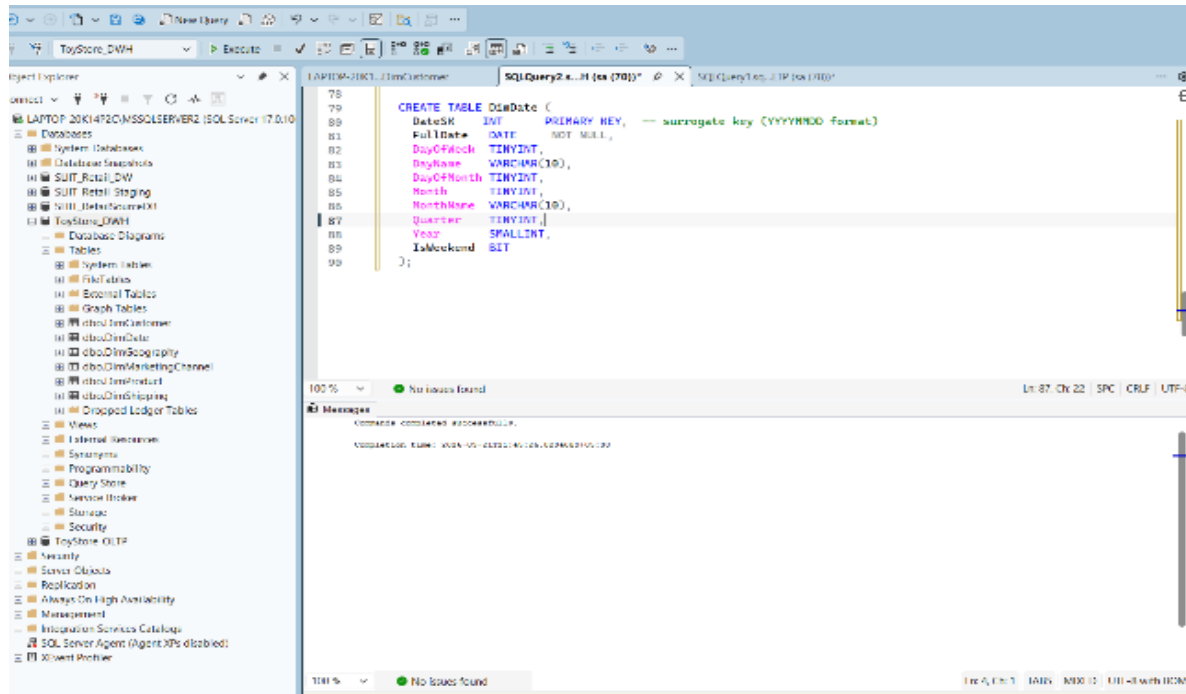
Task 04: Data Warehouse Design & Development

Star Schema was chosen because all dimension tables link directly to the fact table with no further normalisation. This means simpler SQL joins, faster query performance and better compatibility with SSAS cube design.

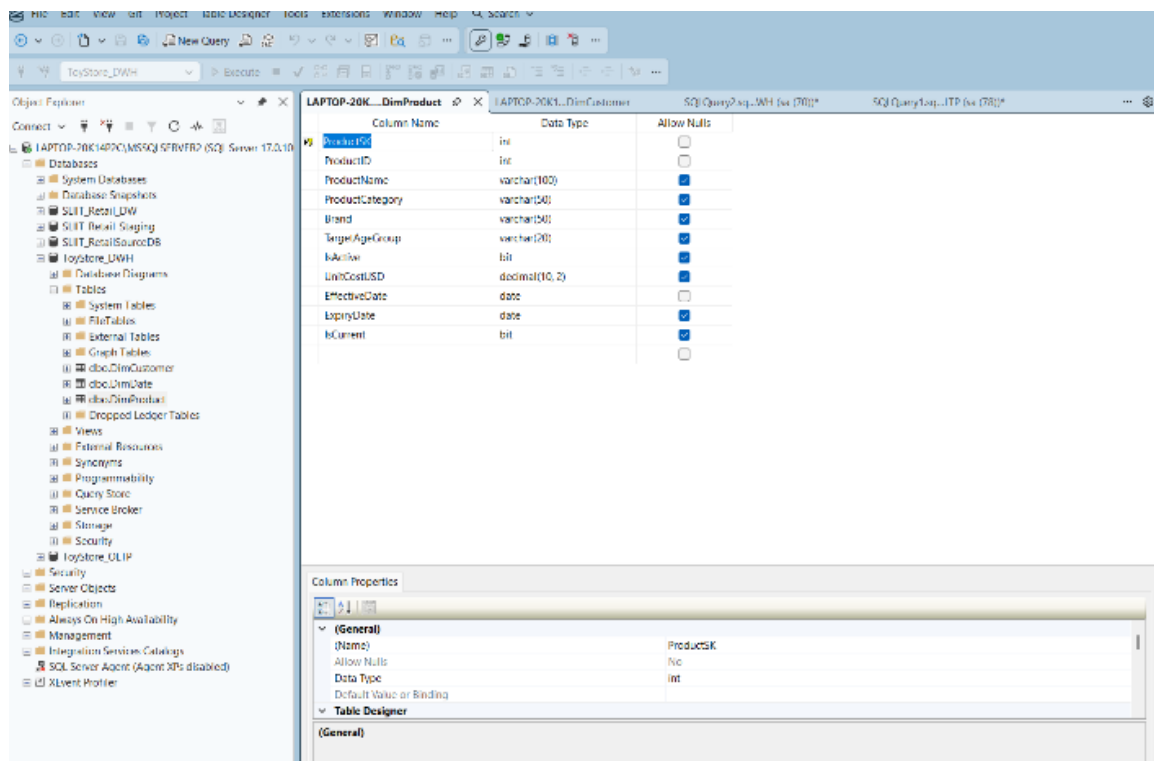
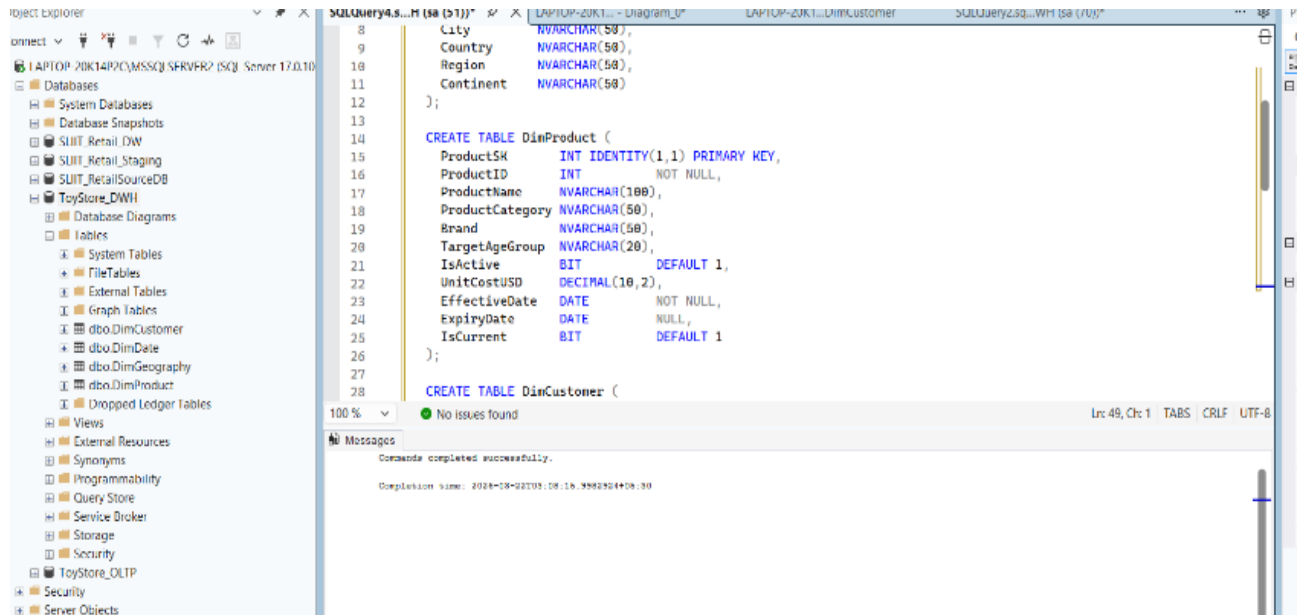
First create ToyStore_DWH.



Create DimDate table



Create DimProduct table



Create DimCustomer table

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'LAPTOP-20K14P2C\MSSQLSERVER2 (SQL Server 17.0.10)'. The 'Tables' folder is expanded, showing the 'dbo' schema. The 'DimCustomer' table is highlighted. The main pane displays the SQL script for creating the 'DimCustomer' and 'DimMarketingChannel' tables. The 'DimCustomer' table has the following columns: CustomerSK (INT, PRIMARY KEY, NOT NULL), UserID (INT, NOT NULL), Email (NVARCHAR(200)), PreferredDevice (NVARCHAR(20)), Country (NVARCHAR(50)), HasPurchased (BIT), RegistrationDate (DATE), EffectiveDate (DATE, NOT NULL), ExpiryDate (DATE, NULL), and IsCurrent (BIT, DEFAULT 1). The 'DimMarketingChannel' table has the following columns: ChannelSK (INT, PRIMARY KEY, NOT NULL) and UTMSource (NVARCHAR(50)). The Messages pane at the bottom shows the command completed successfully.

```
23  EffectiveDate  DATE          NOT NULL,
24  ExpiryDate    DATE          NULL,
25  IsCurrent      BIT            DEFAULT 1
26  );
27
28  CREATE TABLE DimCustomer (
29  CustomerSK    INT IDENTITY(1,1) PRIMARY KEY,
30  UserID        INT            NOT NULL,
31  Email         NVARCHAR(200),
32  PreferredDevice NVARCHAR(20),
33  Country       NVARCHAR(50),
34  HasPurchased   BIT,
35  RegistrationDate DATE,
36  EffectiveDate  DATE          NOT NULL,
37  ExpiryDate     DATE          NULL,
38  IsCurrent      BIT            DEFAULT 1
39  );
40
41  CREATE TABLE DimMarketingChannel (
42  ChannelSK     INT IDENTITY(1,1) PRIMARY KEY,
43  UTMSource      NVARCHAR(50),
44  );
```

The screenshot shows the SQL Server Enterprise Manager interface with the 'Table Designer' for the 'DimCustomer' table. The 'Columns' tab is selected, showing a list of columns with their data types and 'Allow Nulls' property. The 'CustomerSK' column is highlighted. The 'Column Properties' pane at the bottom shows the properties for the 'CustomerSK' column, including its name, data type, and 'Allow Nulls' property.

Column Name	Data Type	Allow Nulls
CustomerSK	int	☐
UserID	int	☐
Email	varchar(200)	☒
PreferredDevice	varchar(20)	☒
Country	varchar(50)	☒
HasPurchased	bit	☒
RegistrationDate	date	☒
EffectiveDate	date	☐
ExpiryDate	date	☒
IsCurrent	bit	☒

Column Properties

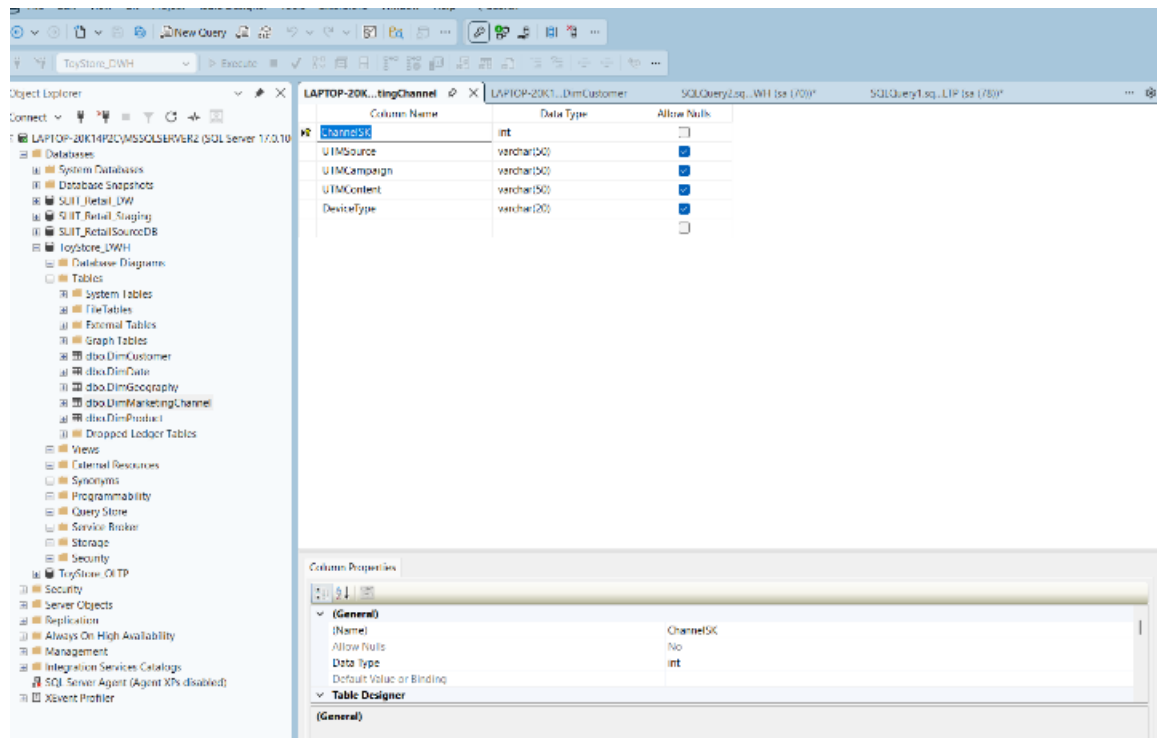
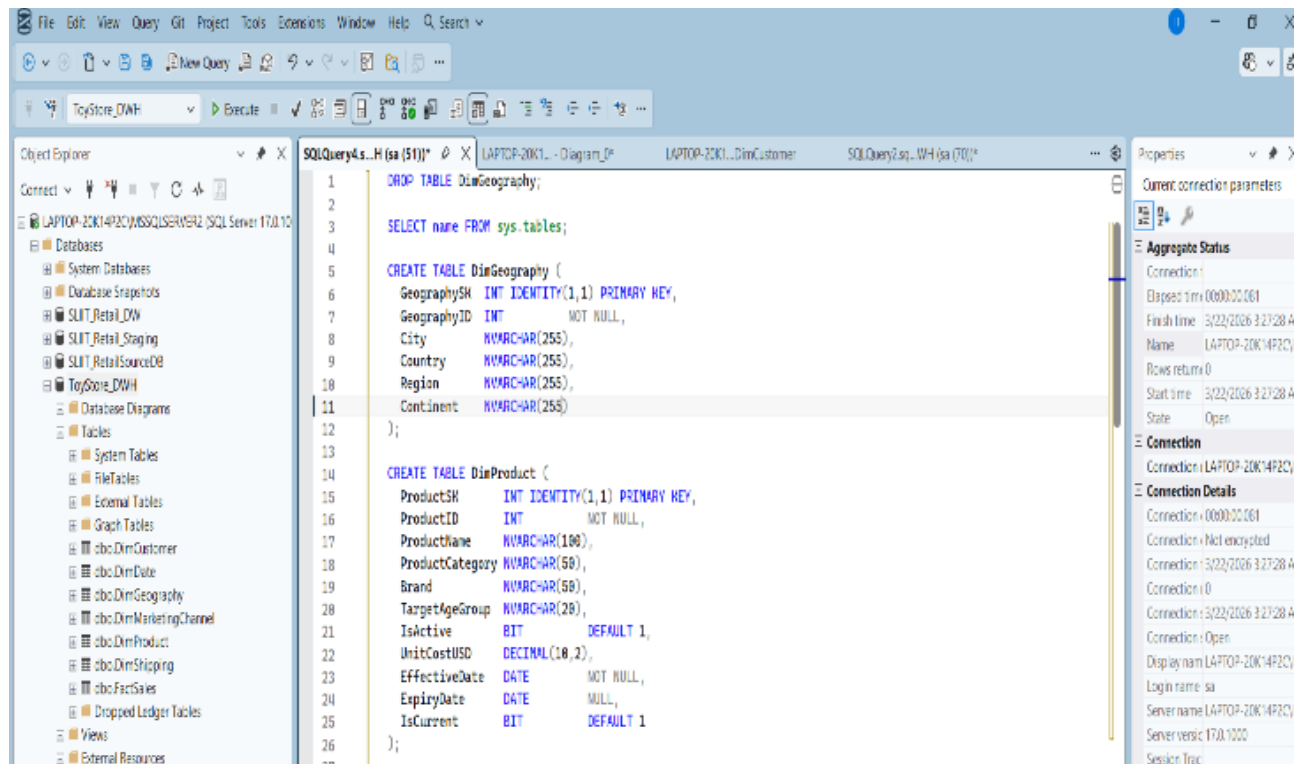
(General)

Property	Value
Name	CustomerSK
Allow Nulls	No
Data Type	int
Default Value or Binding	

Table Designer

(General)

Create DimGeography table



The screenshot displays the SQL Server Enterprise Manager interface. On the left, the 'Object Explorer' shows the server 'LAPTOP-20K14P2C\MSSQLSERVER2 (SQL Server 17.0.10)'. The 'Database Diagrams' folder is expanded, showing a 'Tables' folder. The 'Tables' folder is further expanded, showing a list of tables including 'System Tables', 'FileTables', 'External Tables', 'Graph Tables', 'dbo.DimCustomer', 'dbo.DimDate', 'dbo.DimGeography', 'dbo.DimProduct', 'Dropped Ledger Tables', 'Views', 'External Resources', 'Synonyms', 'Programmability', 'Query Store', 'Service Broker', 'Storage', 'Security', and 'ToyStore_OLTP'.

The right pane shows the 'SQL Query Editor' with a query titled 'SQLQuery4.s...H (sa (51))'. The query is a CREATE TABLE statement for 'DimMarketingChannel'. The query text is as follows:

```

35 RegistrationDate DATE,
36 EffectiveDate DATE NOT NULL,
37 ExpiryDate DATE NULL,
38 IsCurrent BIT DEFAULT 1
39 );
40
41 CREATE TABLE DimMarketingChannel (
42 ChannelSK INT IDENTITY(1,1) PRIMARY KEY,
43 UTMSource NVARCHAR(50),
44 UTMCampaign NVARCHAR(50),
45 UTMContent NVARCHAR(50),
46 DeviceType NVARCHAR(28)
47 );
48
49

```

The status bar at the bottom of the SQL Query Editor indicates '100 %' zoom, 'No issues found', and 'Ltr 45, Chr 1 TABS CRLF UTF-8'. The 'Messages' pane shows the following text:

```

Commands completed successfully.
Completion time: 2024-09-23T10:10:10.2062024100:00

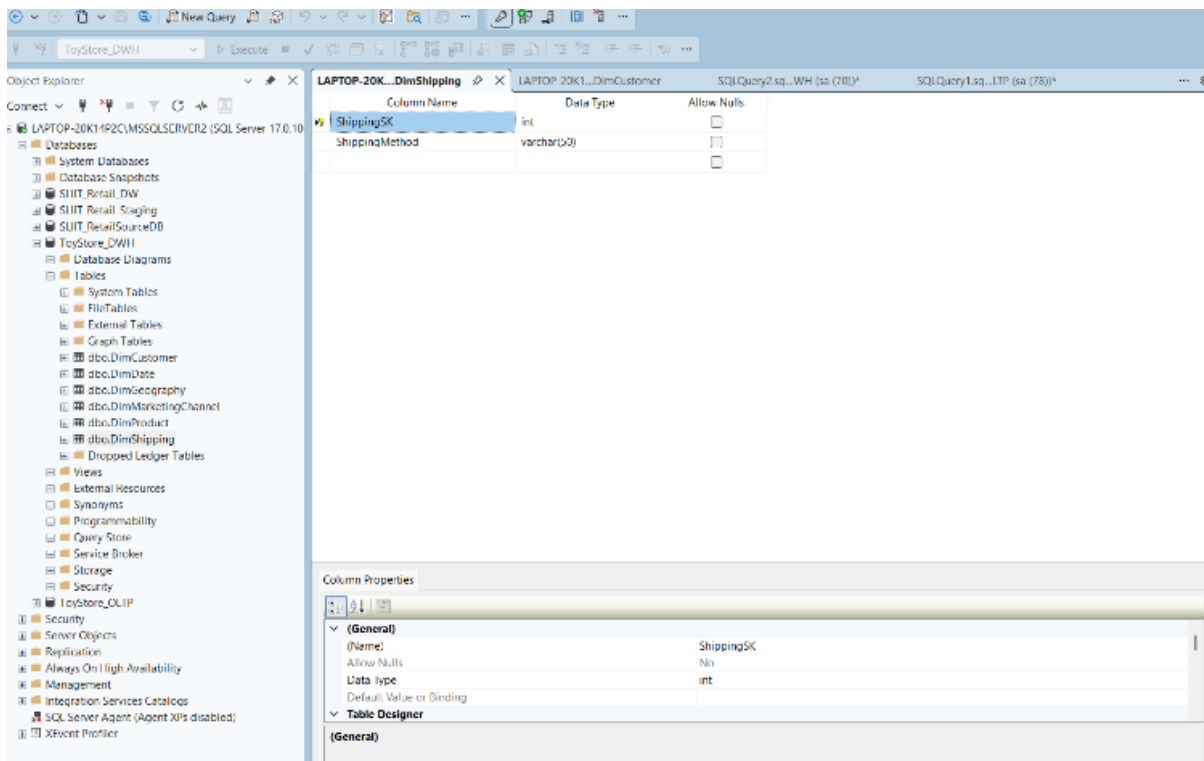
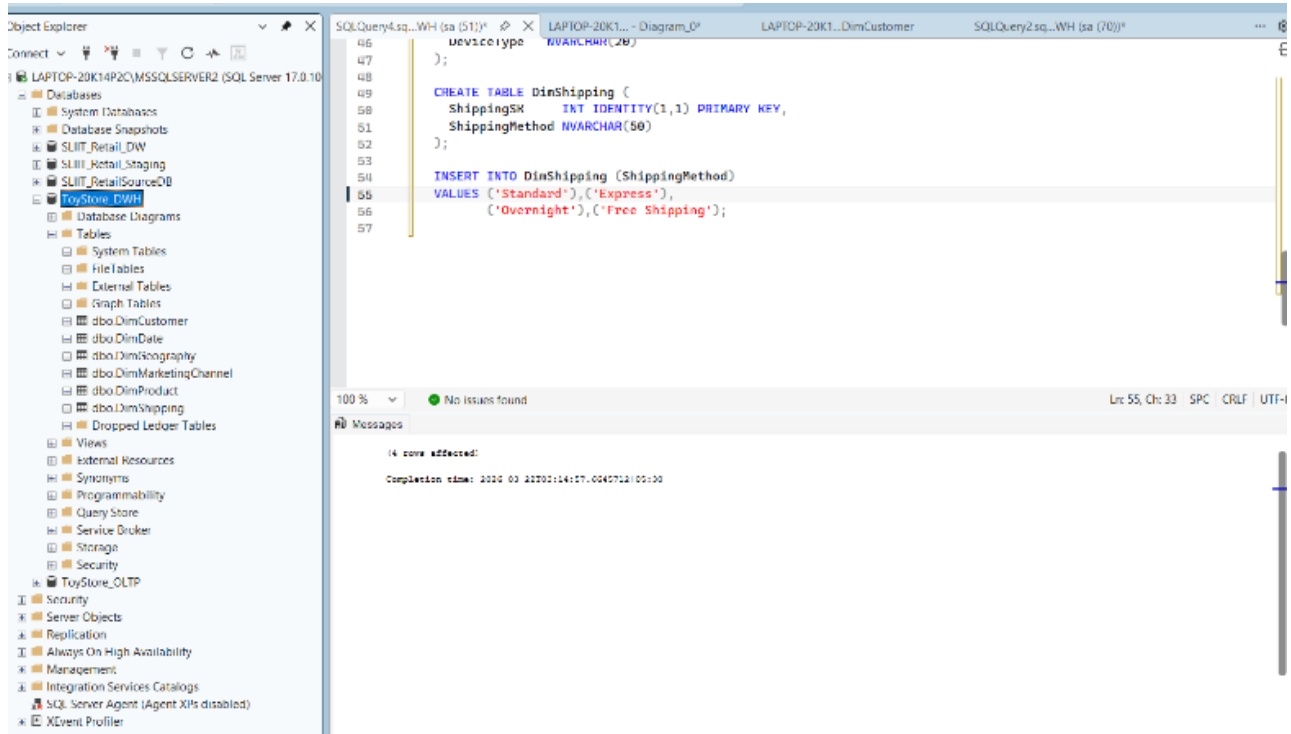
```

The screenshot shows the SQL Server Enterprise Manager interface. The 'Table Designer' for the 'GeographySK' table is open. The table has the following columns:

Column Name	Data Type	Allow Nulls
GeographyID	int	☐
City	varchar(50)	☐
Country	varchar(50)	☒
Region	varchar(50)	☒
Continent	varchar(50)	☒

The 'Table Designer' tab is active, showing the 'General' properties. The 'Name' is 'GeographySK', 'Allow Nulls' is 'No', and 'Data Type' is 'int'. The 'Default Value or Binding' is empty.

Create DimShipping table



Create FactSales table

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Object Explorer' with the 'ToyStore_DWH' database selected. The central pane shows the 'SQLQuery4.s... (sa (51))' window with the following T-SQL script:

```
CREATE TABLE FactSales (  
SalesSK INT IDENTITY(1,1) PRIMARY KEY,  
OrderItemID INT NOT NULL,  
OrderID INT NOT NULL,  
OrderDateSK INT REFERENCES DimDate(DateSK),  
ProductSK INT REFERENCES DimProduct(ProductSK),  
CustomerSK INT REFERENCES DimCustomer(CustomerSK),  
GeographySK INT REFERENCES DimGeography(GeographySK),  
ChannelSK INT REFERENCES DimMarketingChannel(ChannelSK),  
ShippingSK INT REFERENCES DimShipping(ShippingSK),  
RevenueUSD DECIMAL(10,2),  
COGSUSD DECIMAL(10,2),  
DiscountUSD DECIMAL(10,2),  
GrossProfit AS (RevenueUSD - COGSUSD),  
IsRefunded BIT DEFAULT 0,  
RefundAmountUSD DECIMAL(10,2) DEFAULT 0,  
acctm_txn_create_time DATETIME,  
acctm_txn_complete_time DATETIME,  
txn_process_time_hours AS DATEDIFF(HOUR,  
acctm_txn_create_time,  
acctm_txn_complete_time)  
);
```

The right pane shows the 'Properties' window for the connection, displaying details such as 'Connection 1', 'Elapsed time: 00:00:00.014', 'Finish time: 3/22/2016 3:16:40 AM', 'Name: LAPTOP-20K14P2CJM', 'Rows returned: 0', 'Start time: 3/22/2016 3:16:39 AM', 'State: Open', 'Connection (LAPTOP-20K14P2CJM)', 'Connection ID: 00:00:00.074', 'Connection (Not encrypted)', 'Connection 1: 3/22/2016 3:16:39 AM', 'Connection: Open', 'Display name: LAPTOP-20K14P2CJM', 'Login name: sa', 'Server name: LAPTOP-20K14P2CJM', 'Server version: 17.0.1000', 'Session trace: SPID: 51', 'TDS protocol: 6x74000004'.

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Object Explorer' with the 'ToyStore_DWH' database selected. The central pane shows the 'SQLQuery4.s... (sa (51))' window with the following T-SQL script:

```
CREATE TABLE FactSales (  
SalesSK INT IDENTITY(1,1) PRIMARY KEY,  
OrderItemID INT NOT NULL,  
OrderID INT NOT NULL,  
OrderDateSK INT REFERENCES DimDate(DateSK),  
ProductSK INT REFERENCES DimProduct(ProductSK),  
CustomerSK INT REFERENCES DimCustomer(CustomerSK),  
GeographySK INT REFERENCES DimGeography(GeographySK),  
ChannelSK INT REFERENCES DimMarketingChannel(ChannelSK),  
ShippingSK INT REFERENCES DimShipping(ShippingSK),  
RevenueUSD DECIMAL(10,2),  
COGSUSD DECIMAL(10,2),  
DiscountUSD DECIMAL(10,2),  
GrossProfit AS (RevenueUSD - COGSUSD),  
IsRefunded BIT DEFAULT 0,  
RefundAmountUSD DECIMAL(10,2) DEFAULT 0,  
acctm_txn_create_time DATETIME,  
acctm_txn_complete_time DATETIME,  
txn_process_time_hours AS DATEDIFF(HOUR,  
acctm_txn_create_time,  
acctm_txn_complete_time)  
);
```

The right pane shows the 'Properties' window for the connection, displaying details such as 'Connection 1', 'Elapsed time: 00:00:00.014', 'Finish time: 3/22/2016 3:16:40 AM', 'Name: LAPTOP-20K14P2CJM', 'Rows returned: 0', 'Start time: 3/22/2016 3:16:39 AM', 'State: Open', 'Connection (LAPTOP-20K14P2CJM)', 'Connection ID: 00:00:00.074', 'Connection (Not encrypted)', 'Connection 1: 3/22/2016 3:16:39 AM', 'Connection: Open', 'Display name: LAPTOP-20K14P2CJM', 'Login name: sa', 'Server name: LAPTOP-20K14P2CJM', 'Server version: 17.0.1000', 'Session trace: SPID: 51', 'TDS protocol: 6x74000004'.

Populated DimDate

The screenshot displays the SQL Server Enterprise Manager interface. The central pane shows a T-SQL query to populate the DimDate table. The query uses a WHILE loop to insert dates from 2012-01-01 to 2019-12-31. The right pane shows the Properties window for the connection.

```
126 DECLARE @d DATE = '2012-01-01';
127 WHILE @d <= '2019-12-31'
128 BEGIN
129     INSERT INTO DimDate VALUES (
130         CONVERT(INT, FORMAT(@d, 'yyyyMMdd')),
131         @d,
132         DATETIME2(MICROSECOND, @d),
133         DATETIME2(MICROSECOND, @d),
134         DATETIME2(MICROSECOND, @d),
135         MONTH(@d),
136         YEAR(@d),
137         DATETIME2(MICROSECOND, @d),
138         DATETIME2(MICROSECOND, @d),
139         YEAR(@d),
140         CASE WHEN DATETIME2(MICROSECOND, @d)
141             IN (1, 2) THEN 1 ELSE 0 END
142     );
143     SET @d = DATEADD(DAY, 1, @d);
144 END
145
146
147 SELECT 'DimDate' AS TableName,
148        COUNT(*) AS Rows FROM DimDate
149 UNION ALL
150 SELECT 'DimProduct', COUNT(*) FROM DimProduct
151 UNION ALL
152 SELECT 'DimCustomer', COUNT(*) FROM DimCustomer
153 UNION ALL
154 SELECT 'DimGeography', COUNT(*) FROM DimGeography
155 UNION ALL
156 SELECT 'DimMarketingChannel', COUNT(*) FROM DimMarketingChannel
157 UNION ALL
158 SELECT 'DimTimezone', COUNT(*) FROM DimTimezone
```

The right pane shows the Properties window for the connection. The Connection Name is LAPTOP-20K1470C\MSSQLSERVER. The Server Name is LAPTOP-20K1470C. The Login Name is sa. The Password is 70. The TDS protocol is 0x7f000001.

Schema Decision

The data warehouse is designed using a Star Schema. All six dimension tables link directly to the central fact table FactSales with no further normalisation between them. Star Schema was chosen over Snowflake Schema because it provides simpler SQL joins, faster query performance for analytical workloads, and better compatibility with SQL Server Analysis Services (SSAS) cube design.

Grain Statement

The grain of FactSales is one row per order line item. Each row represents one product in one order. An order with two products produces two rows in FactSales.

Surrogate Keys

All dimension tables use a surrogate key named using the convention TableNameSK such as ProductSK and CustomerSK. The surrogate key is a system-generated integer that is independent of the original source ID. FactSales stores only surrogate keys as foreign keys, not the original source IDs.

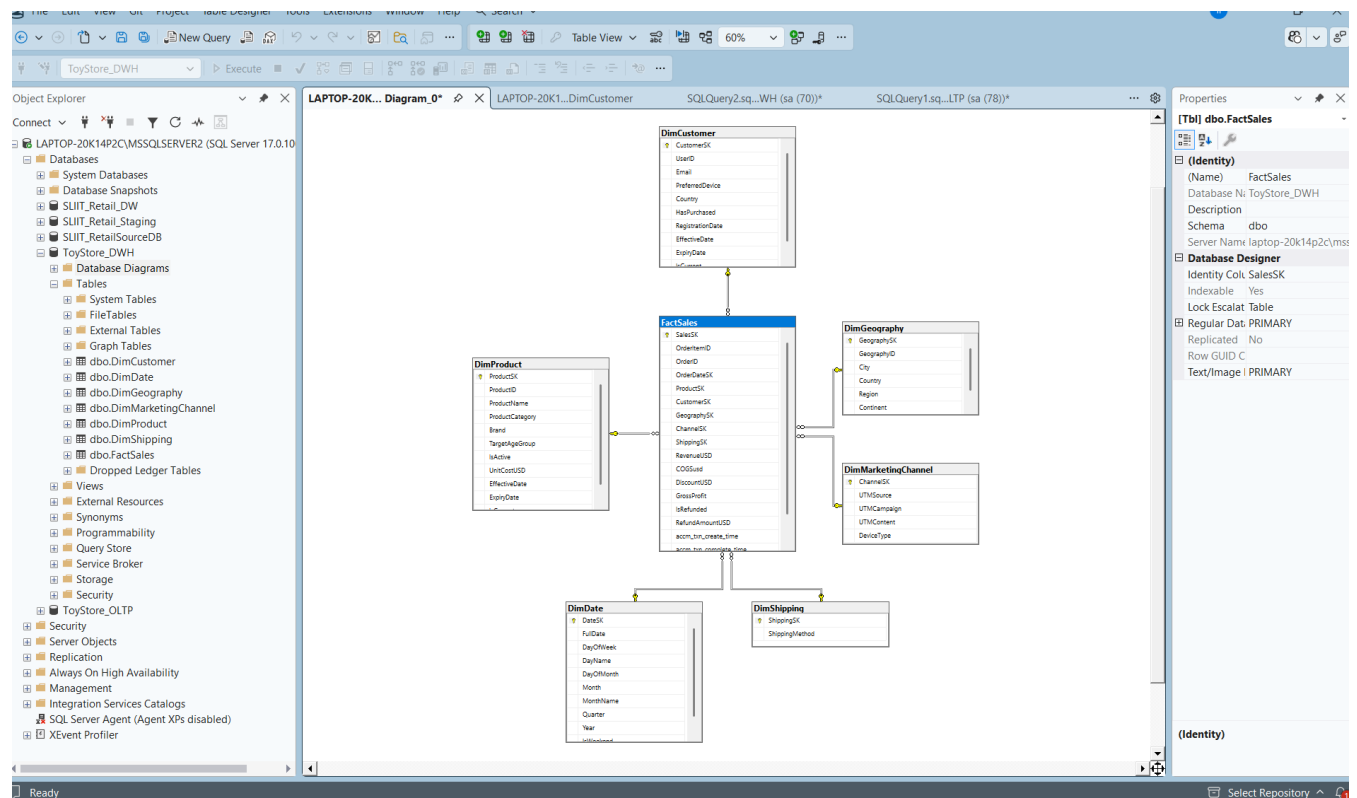
Slowly Changing Dimensions

DimCustomer uses SCD Type 2 to track changes in PreferredDevice and Country over time. When a change is detected, the old row is expired by setting IsCurrent to 0 and ExpiryDate to today, and a new row is inserted with the updated values. DimProduct also uses SCD Type 2 to track changes in UnitCostUSD and TargetAgeGroup.

Computed Columns

GrossProfit and txn_process_time_hours are computed columns. SQL Server calculates them automatically — they are never inserted or updated manually. GrossProfit equals RevenueUSD minus COGSusd. txn_process_time_hours equals the difference in hours between accm_txn_create_time and accm_txn_complete_time.

Star Schema

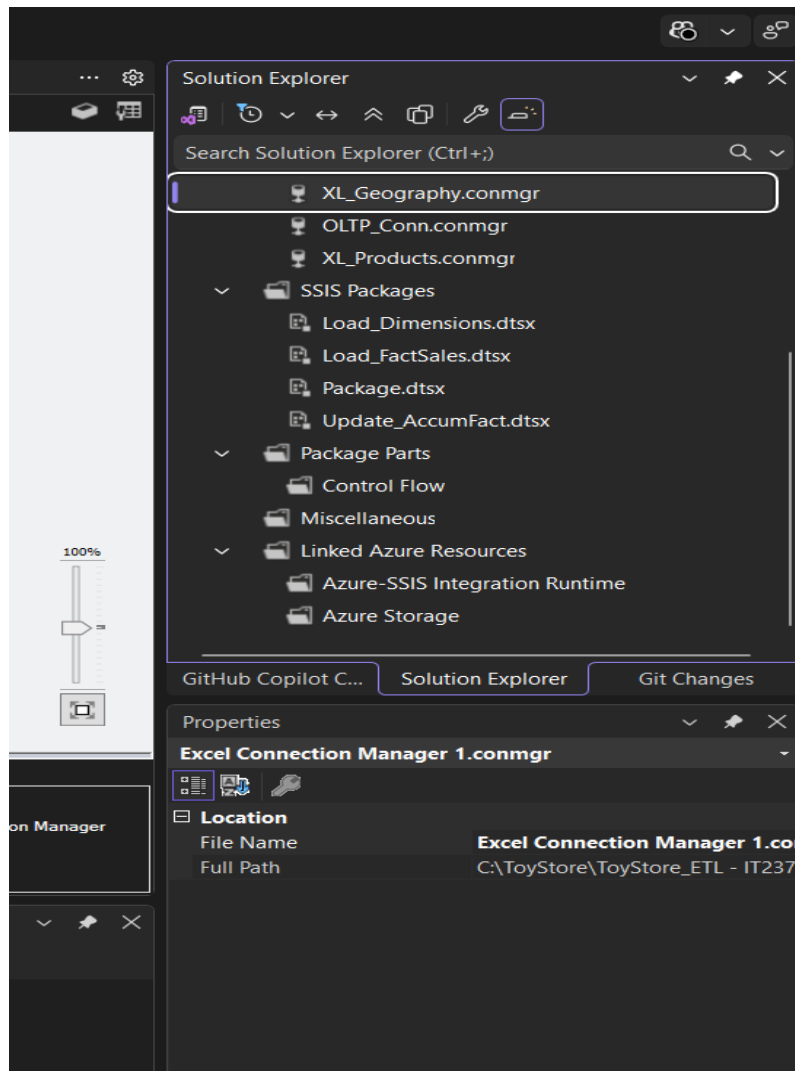


Task 05: ETL Development with SSIS

The ETL process was implemented using SQL Server Integration Services (SSIS) in Visual Studio. Three packages were created to load data from source systems into the ToyStore_DWH data warehouse.

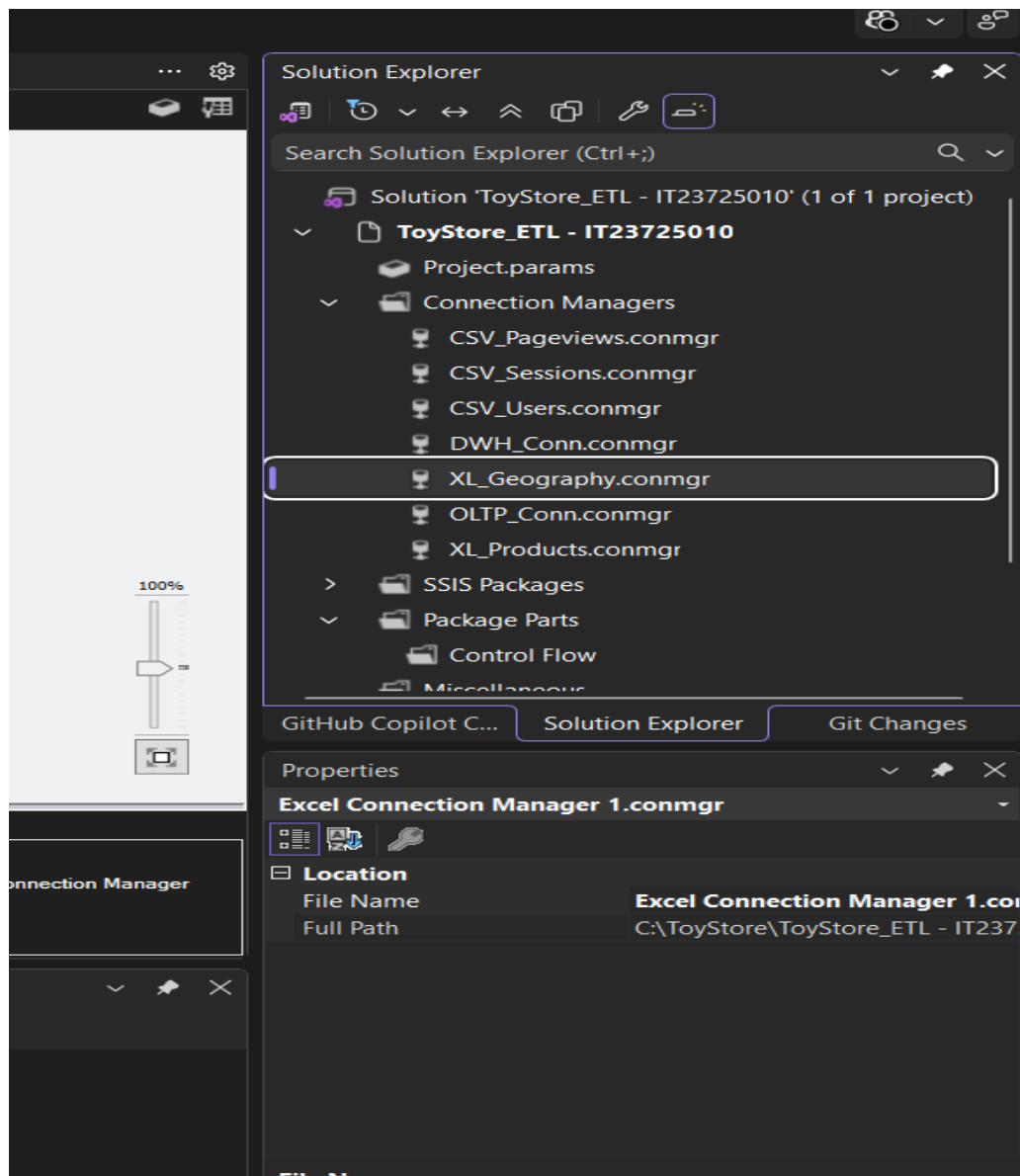
1. Create SSIS Project

1.1 Add 3 Packages



1.2 Create Connection Manager

Seven connection managers were created representing all three source types:

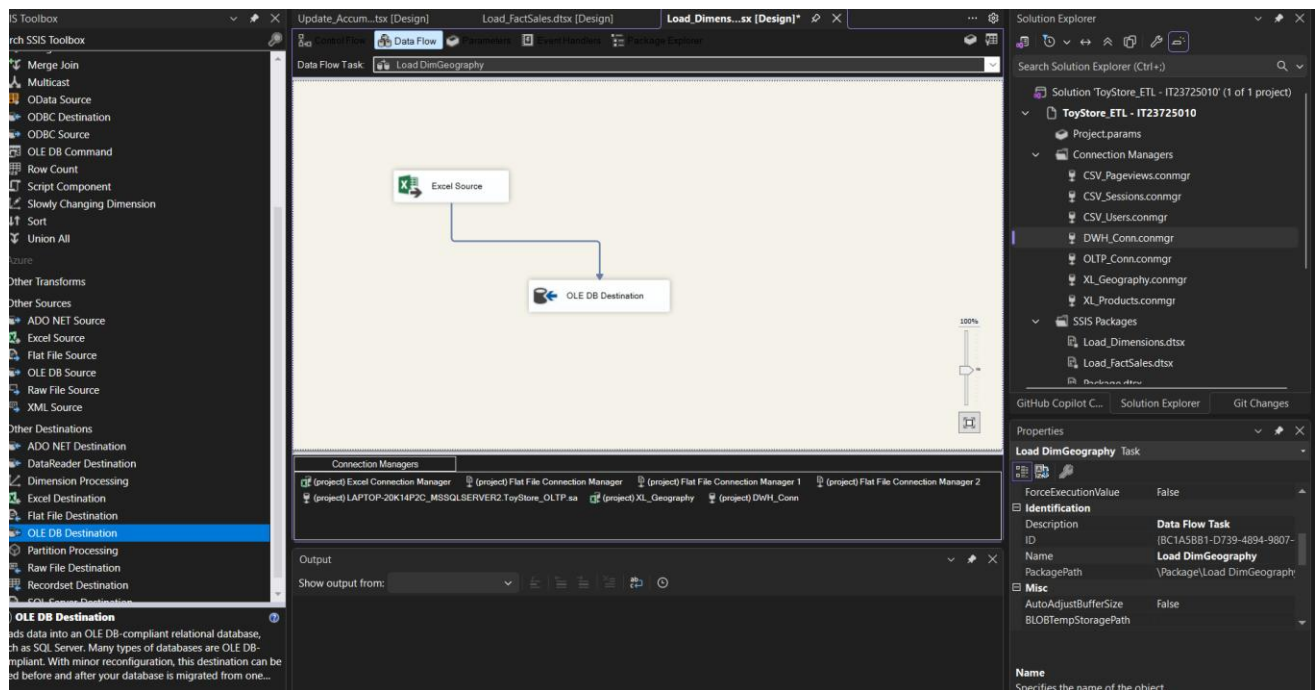


1.3 Package 1: Load_Dimensions.dtsx

Loads all 6 dimension tables from 3 source types in sequence. DimGeography was loaded from geography.xlsx. DimProduct was loaded from products.xlsx with Derived Column adding SCD tracking columns. DimCustomer was loaded from users.csv using a Lookup transformation for SCD Type 2 logic. DimMarketingChannel was loaded from website_sessions.csv using Derived Column, Sort and Aggregate transformations to produce 13 distinct channel combinations.

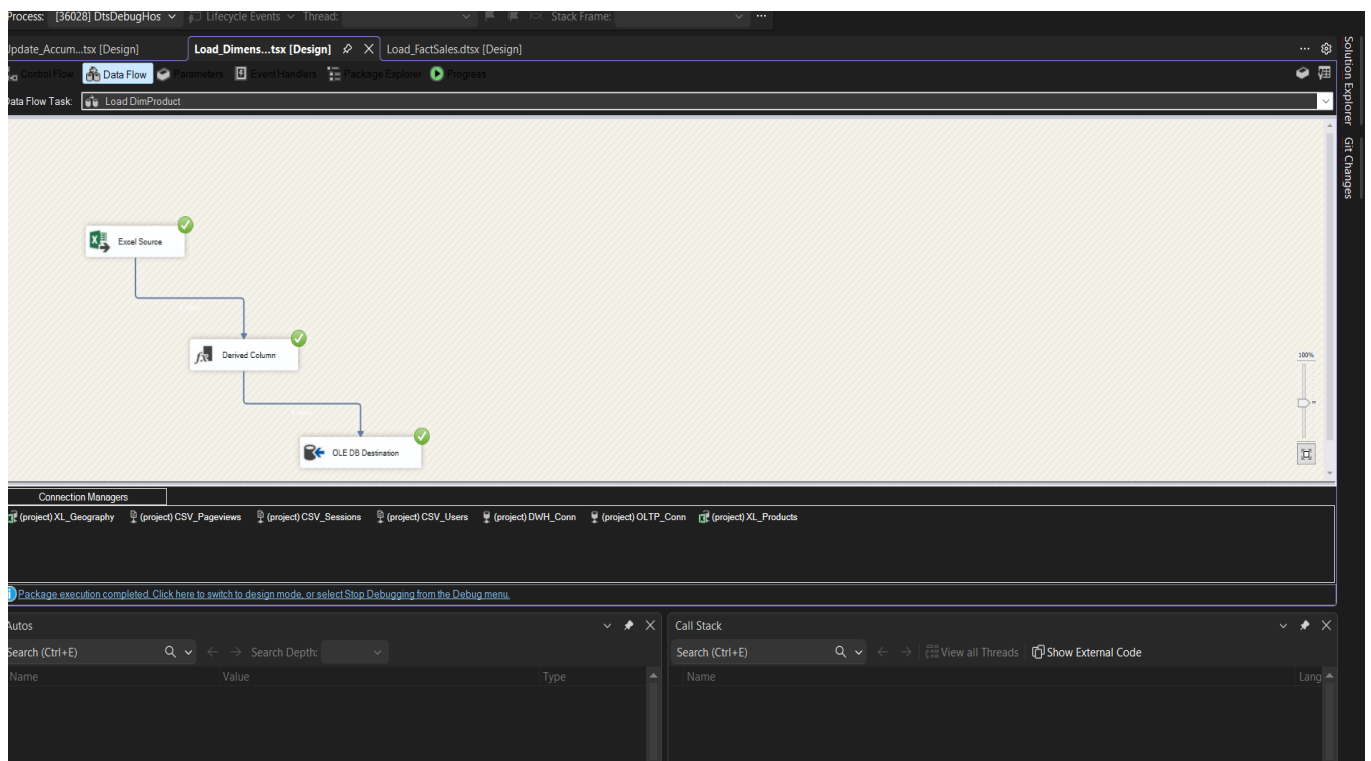
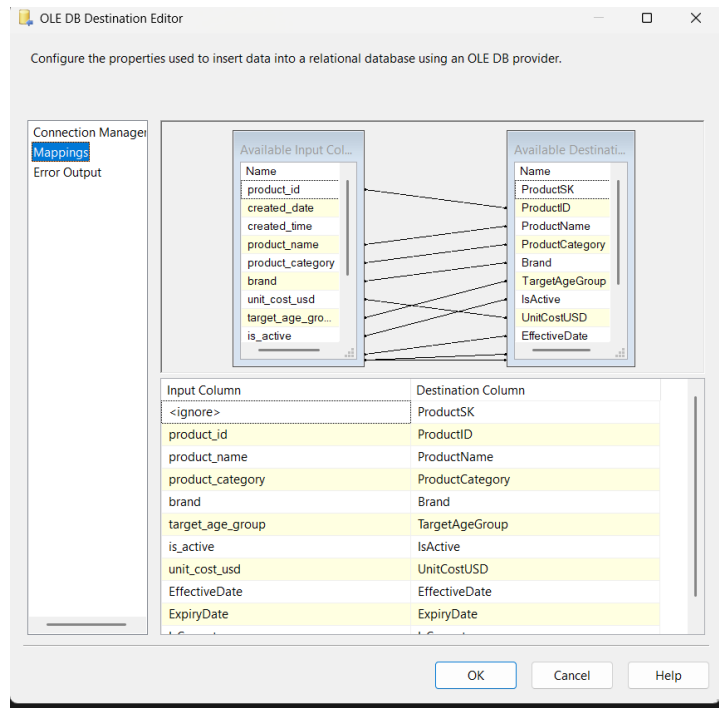
- Data Flow 1 – Load DimGeography

DimGeography was loaded from geography.xlsx using an Excel Source. The http_referer column was excluded. 18 rows were loaded successfully.



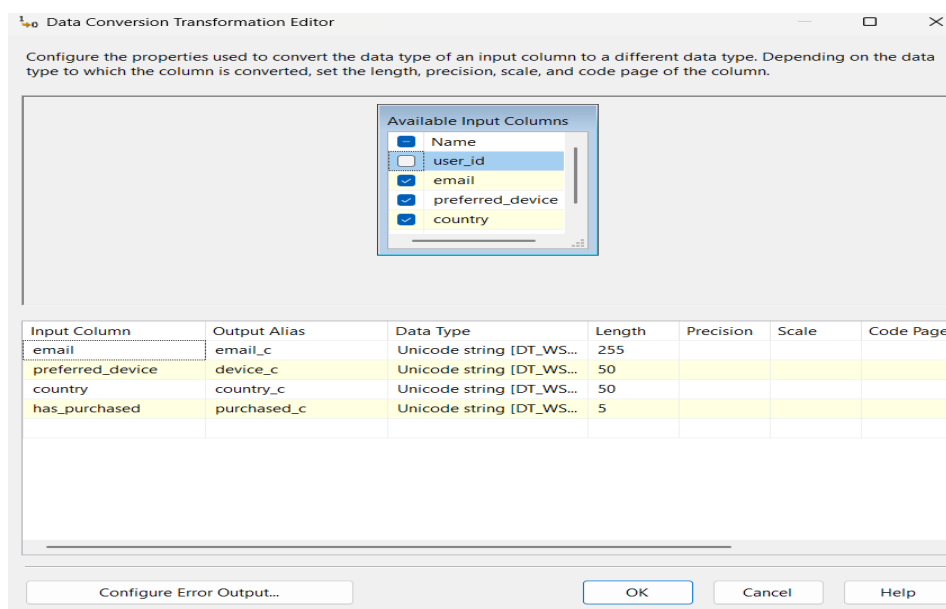
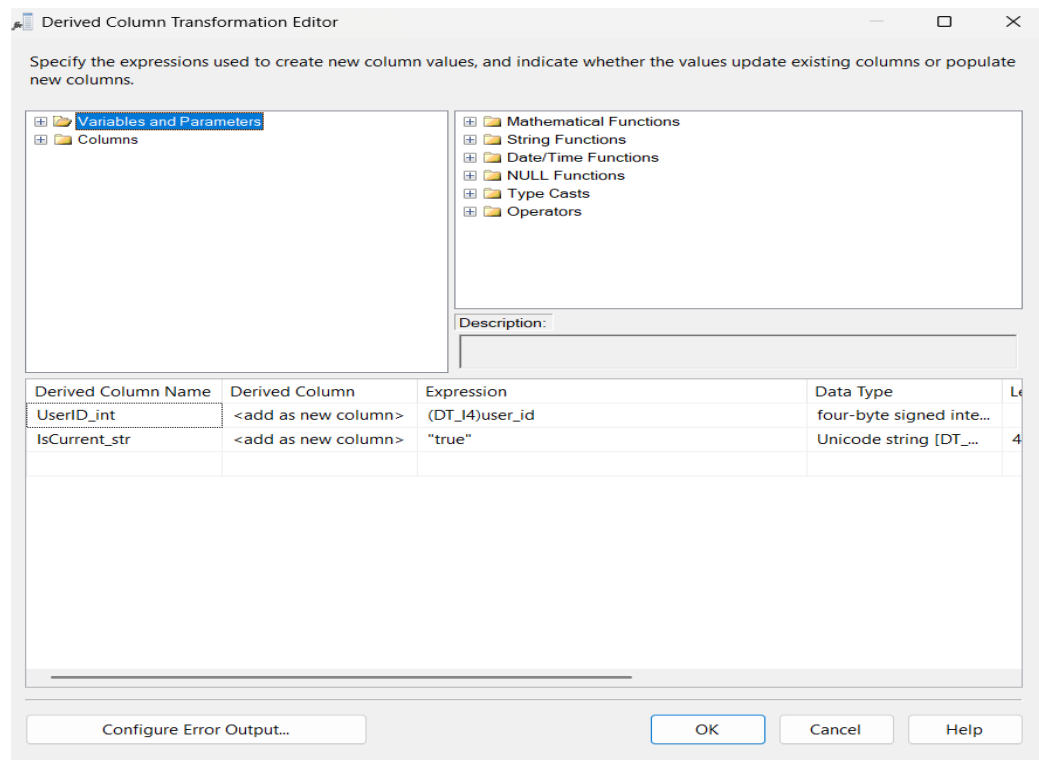
- Data Flow 2 – Load DimProduct

DimProduct was loaded from products.xlsx. A Derived Column transformation added EffectiveDate, ExpiryDate and IsCurrent columns for SCD tracking. 4 rows were loaded successfully.



- Data Flow 3- Load DimCustomer

DimCustomer was loaded from users.csv. A Data Conversion transformation converted Unicode columns to non-unicode format. A Lookup transformation implemented SCD Type 2 logic by checking if each UserID already exists in DimCustomer. New customers were inserted via the No Match output and existing customers were handled via the Match output. 394,318 rows were loaded successfully.



Lookup Transformation Editor

This transform enables the performance of simple equi-joins between the input and a reference data set.

General
Connection
Columns
Advanced
Error Output

Available Input Col...

Available Lookup Columns

Lookup Column	Lookup Operation	Output Alias
CustomerSK	<add as new column>	CustomerSK

OK Cancel Help

OLE DB Destination Editor

Configure the properties used to insert data into a relational database using an OLE DB provider.

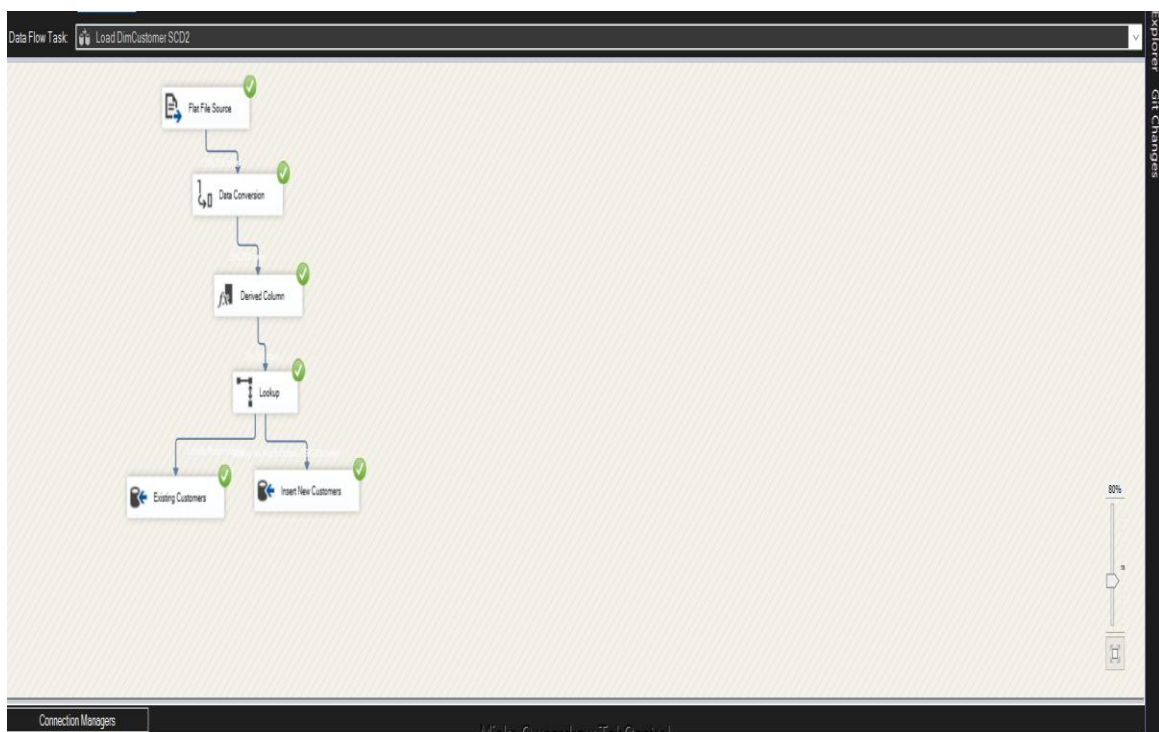
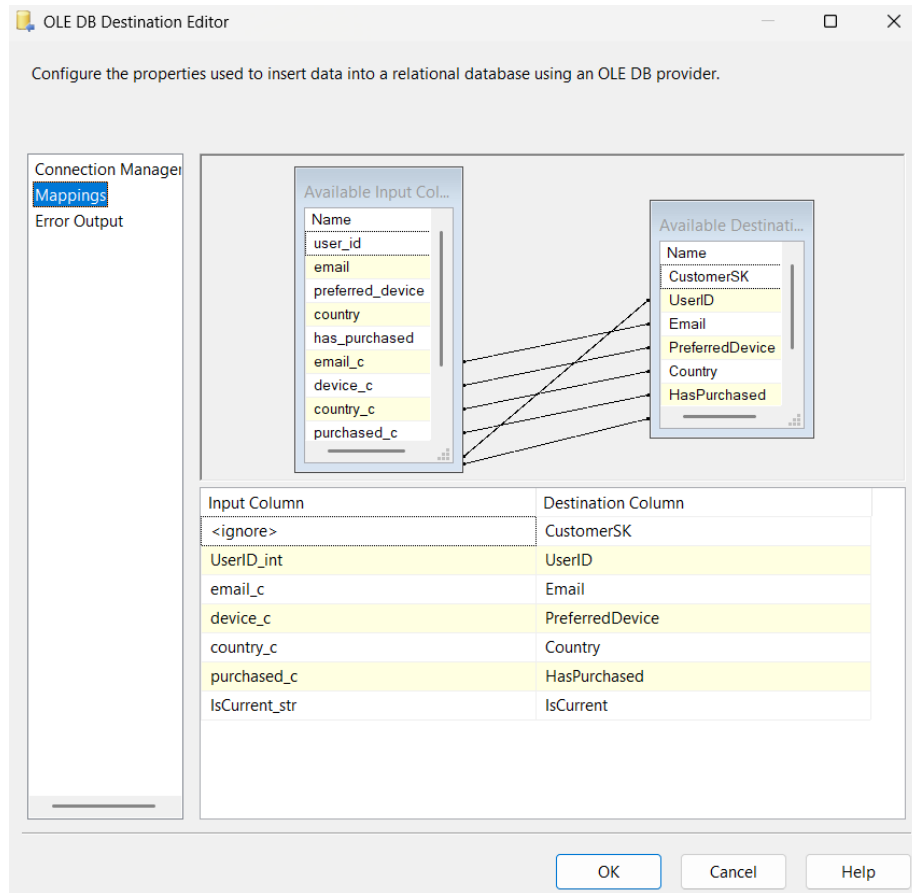
Connection Manager
Mappings
Error Output

Available Input Col...

Available Destinati...

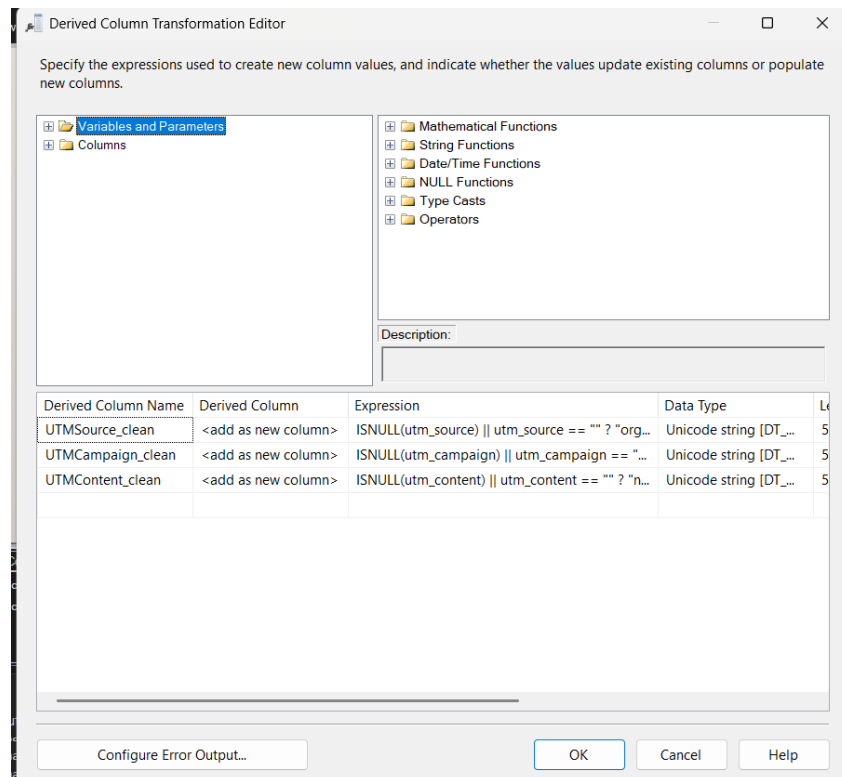
Input Column	Destination Column
<ignore>	CustomerSK
UserID_int	UserID
email_c	Email
device_c	PreferredDevice
country_c	Country
purchased_c	HasPurchased
IsCurrent_str	IsCurrent

OK Cancel Help



- Load DimMarketingChannel

DimMarketingChannel was loaded from website_sessions_clean.csv. A Derived Column transformation replaced NULL UTM values with 'organic' and 'none'. A Sort and Aggregate transformation grouped the data to produce 13 distinct marketing channel combinations. 13 rows were loaded successfully.



Aggregate Transformation Editor

Aggregations

Advanced

Configure the properties used to perform group by operations and to calculate aggregate values. Optionally, apply comparison options to the operation. To configure multiple group by operations, click Advanced.

Advanced

Available Input Columns

☒ Name

☐ (*)

☐ website_session_id

☐ created_date

☐ created_time

☐ user_id

☐ is_repeat_session

Input Column	Output Alias	Operation	Comparison
UTMSource_clean	UTMSource_clean	Group by	
UTMCampaign_clean	UTMCampaign_clean	Group by	
UTMContent_clean	UTMContent_clean	Group by	
device_type	device_type	Group by	

OK

Cancel

Help

Sort Transformation Editor

Specify the columns to sort, and set their sort type and their sort order. All nonselected columns are copied unchanged.

Available Input Columns

☒ Name

☐ website_session_id

☐ created_date

☐ created_time

☐ user_id

☐ is_repeat_session

☐ utm_source

Pass Thr...

☒

☒

☒

☒

☒

☒

☒

Input Column	Output Alias	Sort Type	Sort Order	Condition
device_type	device_type	ascending	4	
UTMSource_clean	UTMSource_clean	ascending	1	
UTMCampaign_clean	UTMCampaign_clean	ascending	2	
UTMContent_clean	UTMContent_clean	ascending	3	

☐ Remove rows with duplicate sort values

OK

Cancel

Help

Process: [32688] DtsDebugHost | Lifecycle Events | Thread: | Stack Frame: |

Load_Dimens...tsx [Design] | Update_Accum...tsx [Design] | Load_FactSales.dtsx [Design]

Data Flow Task: Load DimMarketingChannel

```
graph TD; A[Flat File Source] --> B[Derived Column]; B --> C[Data Conversion]; C --> D[Sort]; D --> E[Aggregate]; E --> F[OLE DB Destination];
```

Properties: Load DimMarketingChannel Task

- Execution
 - DelayValidation: False
 - Disable: False
 - DisableEventHandlers: False
 - FailPackageOnFailure: False
 - FailParentOnFailure: False
 - MaximumErrorCount: 1
- Forced Execution Value
 - ForcedExecutionValue: 0
 - ForcedExecutionValueType: Int32
 - ForceExecutionValue: False
- Identification
 - Description: Data Flow Task
 - ID: {F8B07C98-651D-40C}
 - Name: Load DimMarketingChannel
 - PackagePath: (Package)Load DimMarketingChannel
- Misc
 - [Derived Column].[Derived + utm_campaign] == "" ? 0 : 1
 - [Derived Column].[Derived + utm_content] == "" ? 0 : 1
 - [Derived Column].[Derived + utm_source] == "" ? 0 : 1
 - [Sort].[MaximumThreads]: -1
 - AutoAdjustBufferSize: False
 - BLOBTempStoragePath: BLOBTempStoragePath
 - BufferTempStoragePath: BufferTempStoragePath
 - DefaultBufferMaxRows: 10000
 - DefaultBufferSize: 10485760
 - EngineThreads: 10
 - ExecValueVariable: <none>
- Expressions
 - ForceExecutionResult: None
 - HasExpressions: False
 - LocaleID: English (United States)
 - LoggingMode: UseParentSetting

Autos: Search (Ctrl+E) | Search Depth: |

Call Stack: Search (Ctrl+E) | View all Threads | Show External Code

Connection Managers: Package execution completed. Click here to switch to design mode, or select Stop Debugging from the Debug menu.

Process: [36636] DtsDebugHost | Lifecycle Events | Thread: | Stack Frame: |

Load_Dimens...tsx [Design] | Update_Accum...tsx [Design] | Load_FactSales.dtsx [Design]

Control Flow Task: Load DimMarketingChannel

```
graph TD; A[Load DimGeography] --> B[Load DimProduct]; B --> C[Load DimCustomer SCD2]; C --> D[Load DimMarketingChannel];
```

Properties: Load DimMarketingChannel Task

- Execution
 - DelayValidation: False
 - Disable: False
 - DisableEventHandlers: False
 - FailPackageOnFailure: False
 - FailParentOnFailure: False
 - MaximumErrorCount: 1
- Forced Execution Value
 - ForcedExecutionValue: 0
 - ForcedExecutionValueType: Int32
 - ForceExecutionValue: False
- Identification
 - Description: Data Flow Task
 - ID: {39F6158D-0EB0-43DF-8E2F}
 - Name: Load DimMarketingChannel
 - PackagePath: (Package)Load DimMarketingChannel
- Misc
 - [Derived Column].[Derived + ISNULL(utm_campaign) || utm_campaign]
 - [Derived Column].[Derived + ISNULL(utm_content) || utm_content]
 - [Derived Column].[Derived + ISNULL(utm_source) || utm_source]
 - [Sort].[MaximumThreads]: -1
 - AutoAdjustBufferSize: False
 - BLOBTempStoragePath: BLOBTempStoragePath
 - BufferTempStoragePath: BufferTempStoragePath
 - DefaultBufferMaxRows: 10000
 - DefaultBufferSize: 10485760
 - EngineThreads: 10
 - ExecValueVariable: <none>
- Expressions
 - ForceExecutionResult: None
 - HasExpressions: False
 - LocaleID: English (United States)
 - LoggingMode: UseParentSetting

Autos: Search (Ctrl+E) | Search Depth: |

Call Stack: Search (Ctrl+E) | View all Threads | Show External Code

Connection Managers: (project) CSV_Pageviews | (project) CSV_Sessions | (project) CSV_Users | (project) DWH_Conn | (project) OLTP_Conn | (project) XL_Geography | (project) XL_Products

Package execution completed with success. Click here to switch to design mode, or select Stop Debugging from the Debug menu.

SQL Server Enterprise Edition (64-bit) - Microsoft SQL Server 17.0.100.60

Server Explorer: LAPTOP-20K14P2C\MSSQLSERVER2 (SQL Server 17.0.100.60)

Properties: Current connection parameters

Aggregate Status: Connection 1, Elapsed time: 00:00:00.298, Finish time: 3/22/2025 10:16:23 PM, Name: LAPTOP-20K14P2C\M, Rows returned: 4, Start time: 3/22/2025 10:16:23 PM, Status: Open

Connection: Connection 1, LAPTOP-20K14P2C\M, Connection ID: 6050106298, Connection ID: Not encrypted, Connection ID: 3/22/2025 10:16:23 PM, Connection ID: 4, Connection ID: 3/22/2025 10:16:23 PM, Connection ID: Open, Display name: LAPTOP-20K14P2C\M, Login name: sa, Server name: LAPTOP-20K14P2C\M, Server version: 17.0.100.60, Spooler: True, SPID: 78, TDS protocol: 6474000004

Name: The name of the connection.

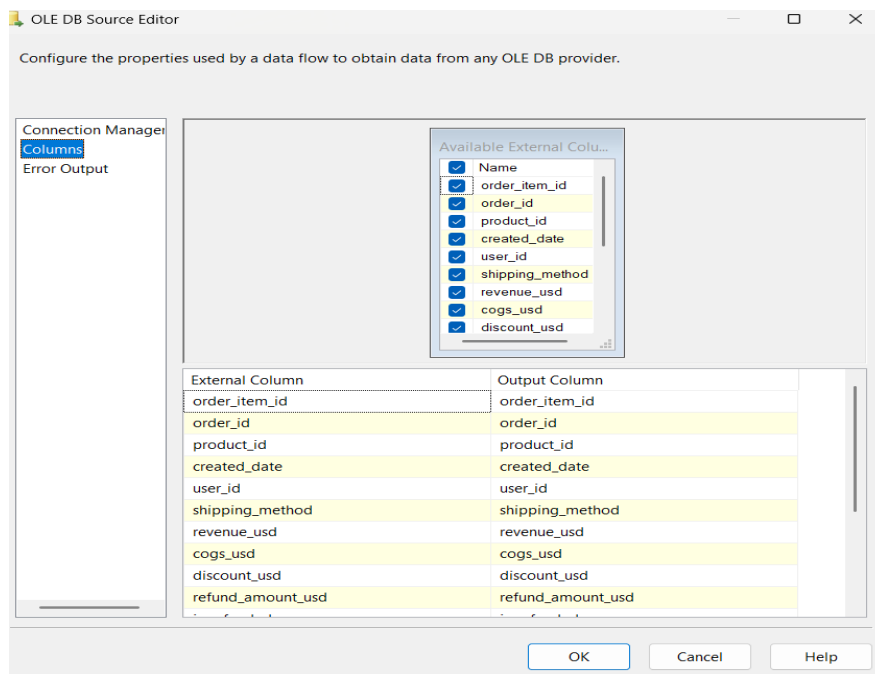
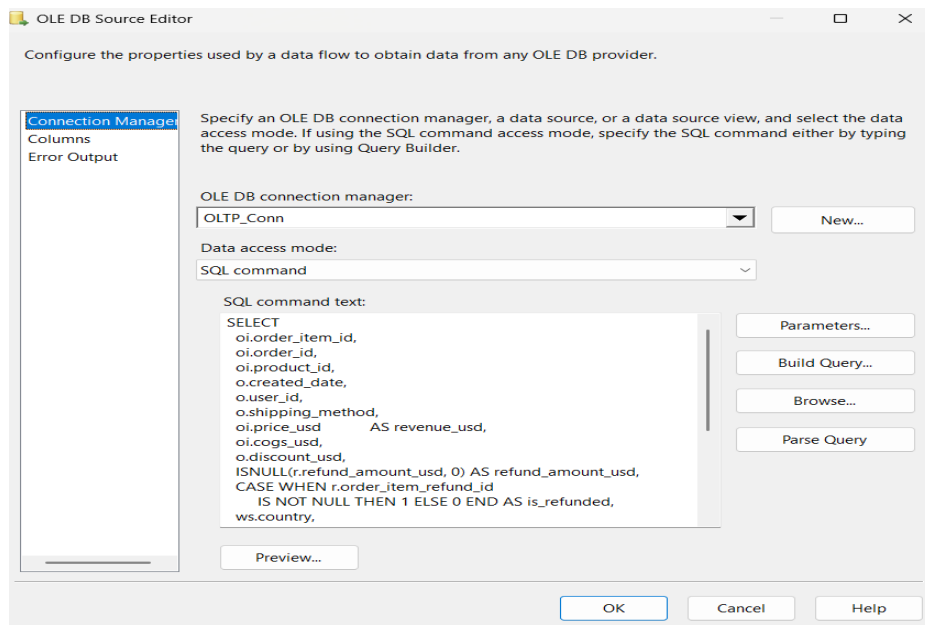
```
7 SELECT 'DisGeography' AS tbl, COUNT(*) AS rows FROM DisGeography
8 UNION ALL
9 SELECT 'DisProduct' AS tbl, COUNT(*) AS rows FROM DisProduct
10 UNION ALL
11 SELECT 'DisCustomer' AS tbl, COUNT(*) AS rows FROM DisCustomer
12 UNION ALL
13 SELECT 'DisMarketingChannel' AS tbl, COUNT(*) AS rows FROM DisMarketingChannel;
```

100% No issues found

tbl	rows
DisGeography	10
DisProduct	4
DisCustomer	304318
DisMarketingChannel	13

Package 1.4- Load_FactSales.dtx

Loads FactSales by joining order_items, orders, order_item_refunds and website_sessions in a single OLE DB Source query. Six Lookup transformations replace natural keys with surrogate keys. A Derived Column adds accm_txn_create_time using GETDATE(). All Lookup No Match outputs are redirected to FactSales_ErrorLog for error tracking.



Lookup DateSK:

Lookup Transformation Editor

This transform enables the performance of simple equi-joins between the input and a reference data set.

General
Connection
Columns
Advanced
Error Output

Specify a data source to use. You can select a table in a data source view, a table in a database connection, or the results of an SQL query.

OLE DB connection manager:
DWH_Conn New...

☐ Use a table or a view:
New...

☒ Use results of an SQL query:
SELECT DateSK, FullDate FROM DimDate
Build Query...
Browse...
Parse Query

Preview...

OK Cancel Help

Lookup Transformation Editor

This transform enables the performance of simple equi-joins between the input and a reference data set.

General
Connection
Columns
Advanced
Error Output

Available Input ...

Name
order_item_id
order_id
product_id
created_date
user_id
shipping_m...
revenue_usd
cogs_usd
discount_usd

Available Lookup Columns

Name	Index
DateSK	
FullDate	

Lookup Column Lookup Operation Output Alias

DateSK	<add as new column>	DateSK
--------	---------------------	--------

OK Cancel Help

Lookup ProductSK:

Lookup Transformation Editor

This transform enables the performance of simple equi-joins between the input and a reference data set.

General
Connection
Columns
Advanced
Error Output

Specify a data source to use. You can select a table in a data source view, a table in a database connection, or the results of an SQL query.

OLE DB connection manager:
DWH_Conn New...

☐ Use a table or a view:
New...

☒ Use results of an SQL query:
SELECT ProductSK, ProductID FROM DimProduct WHERE
IsCurrent = 'true'
Build Query...
Browse...
Parse Query

Preview...

OK Cancel Help

Lookup Transformation Editor

This transform enables the performance of simple equi-joins between the input and a reference data set.

General
Connection
Columns
Advanced
Error Output

Available Input ...

Name
order_item_id
order_id
product_id
created_date
user_id
shipping_m...
revenue_usd
cogs_usd
discount_usd

Available Lookup Columns

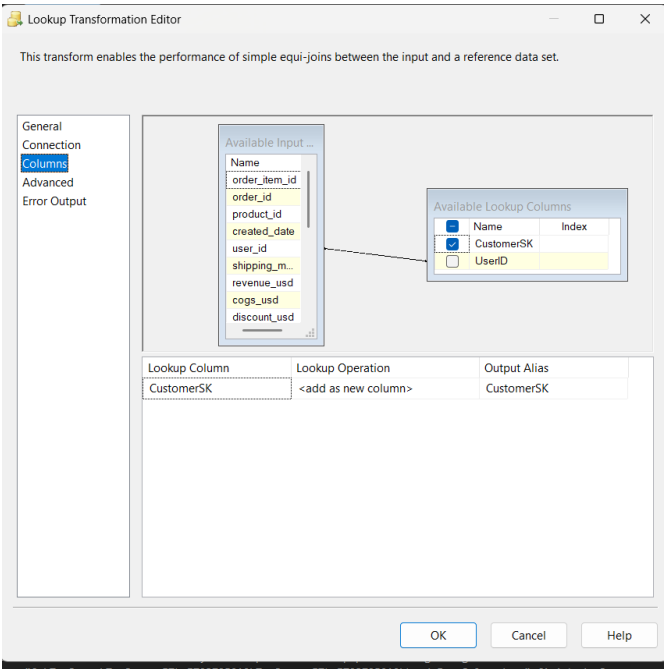
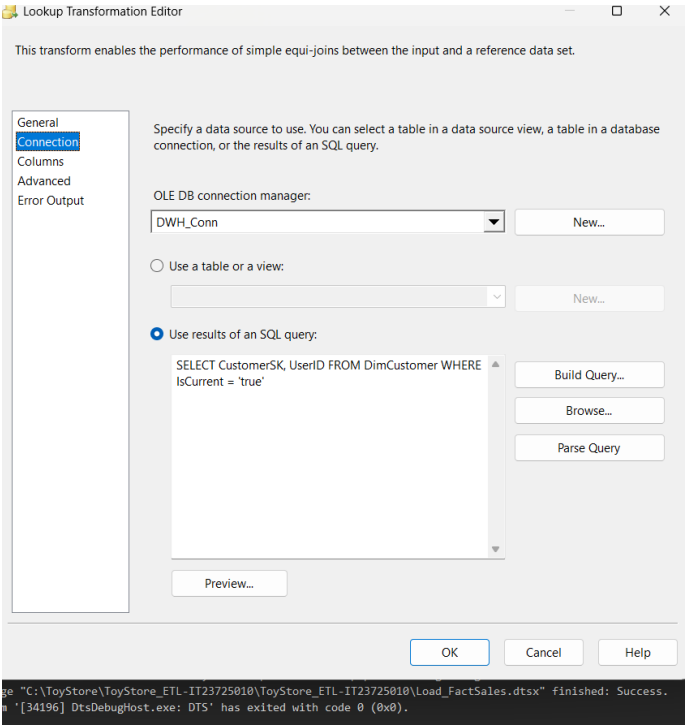
Name	Index
ProductSK	
ProductID	

Lookup Column Lookup Operation Output Alias

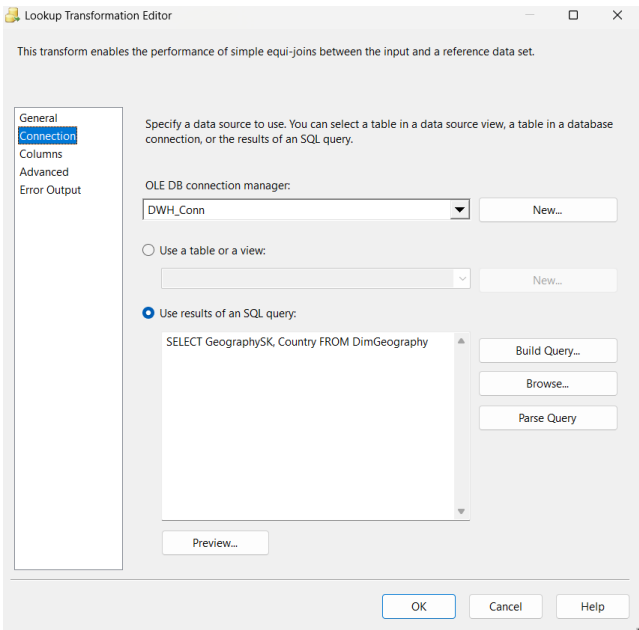
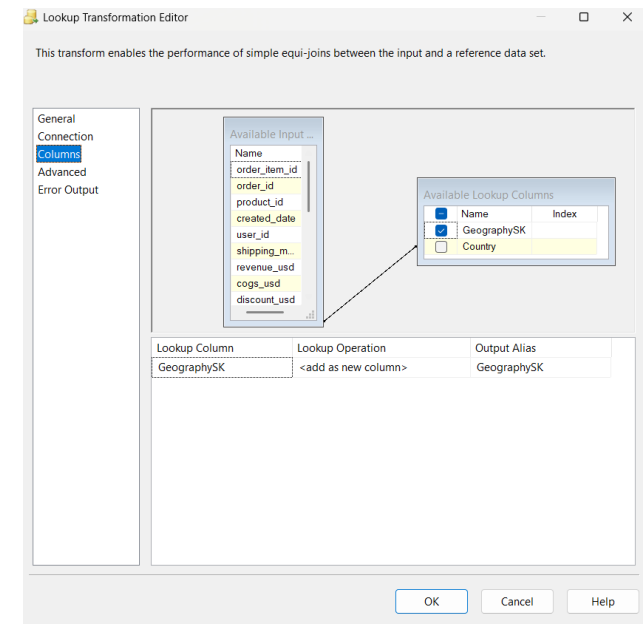
ProductSK	<add as new column>	ProductSK
-----------	---------------------	-----------

OK Cancel Help

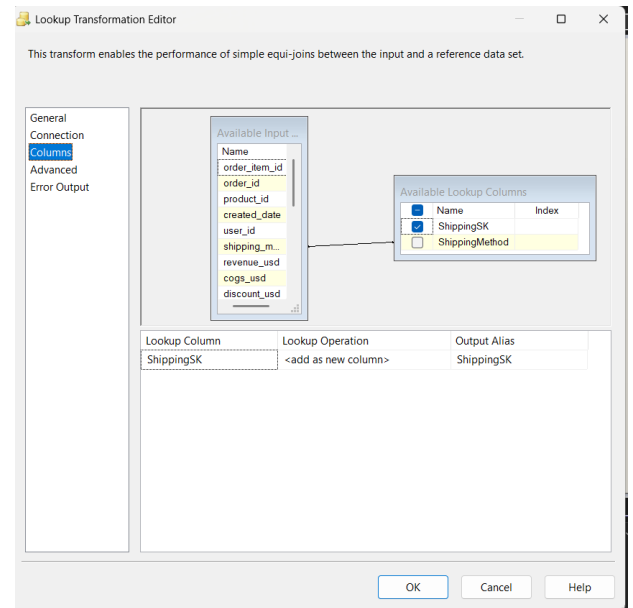
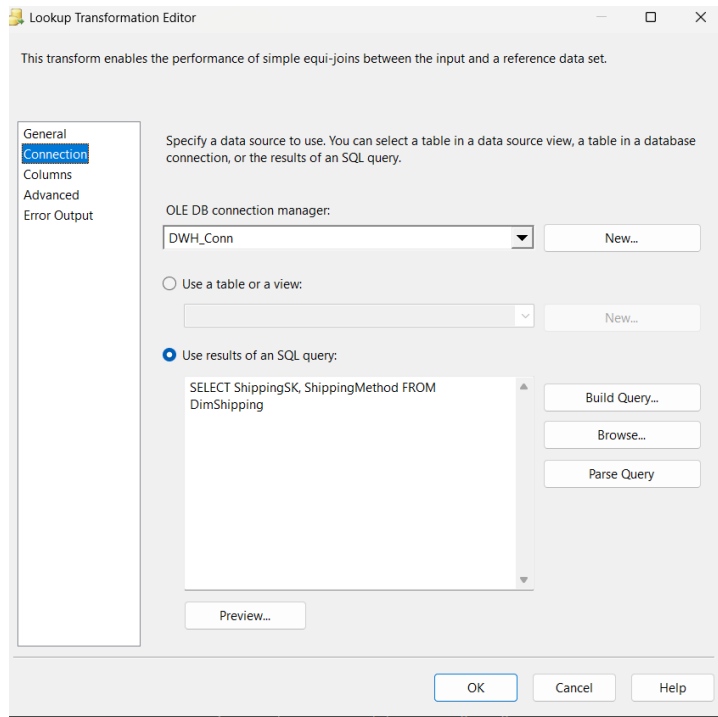
Lookup CustomerSK:



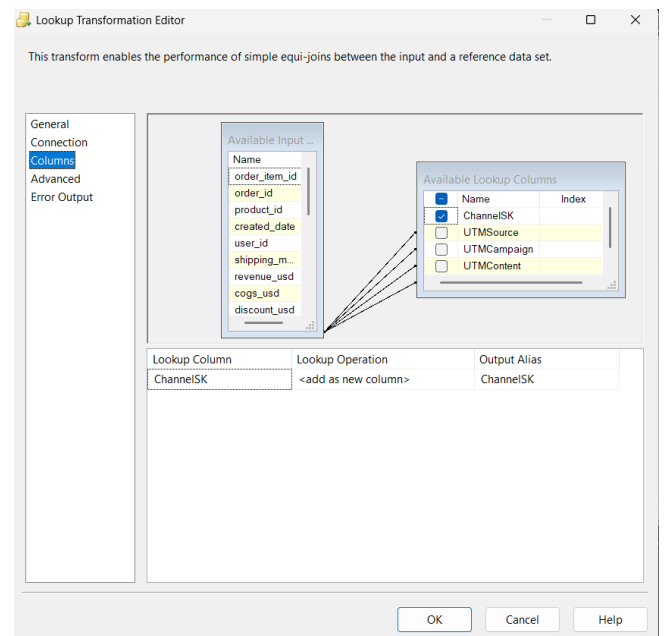
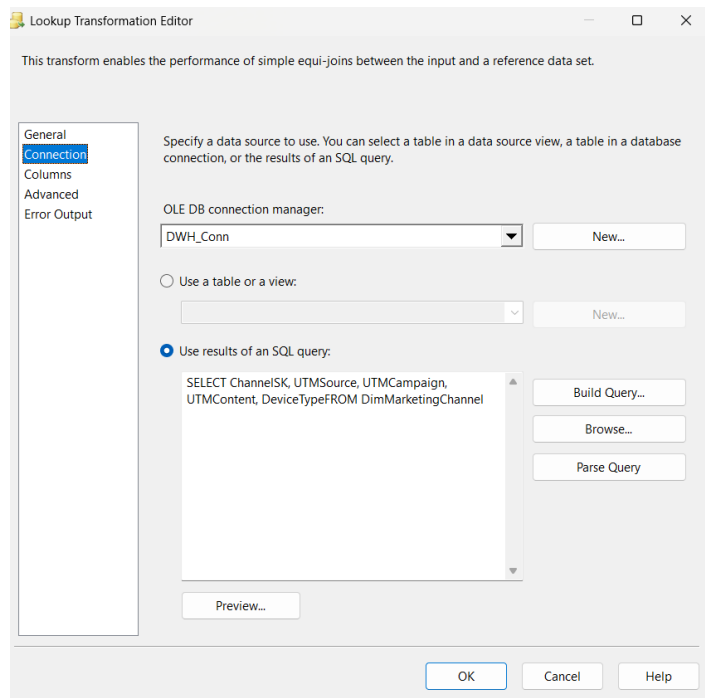
Lookup GeographySK



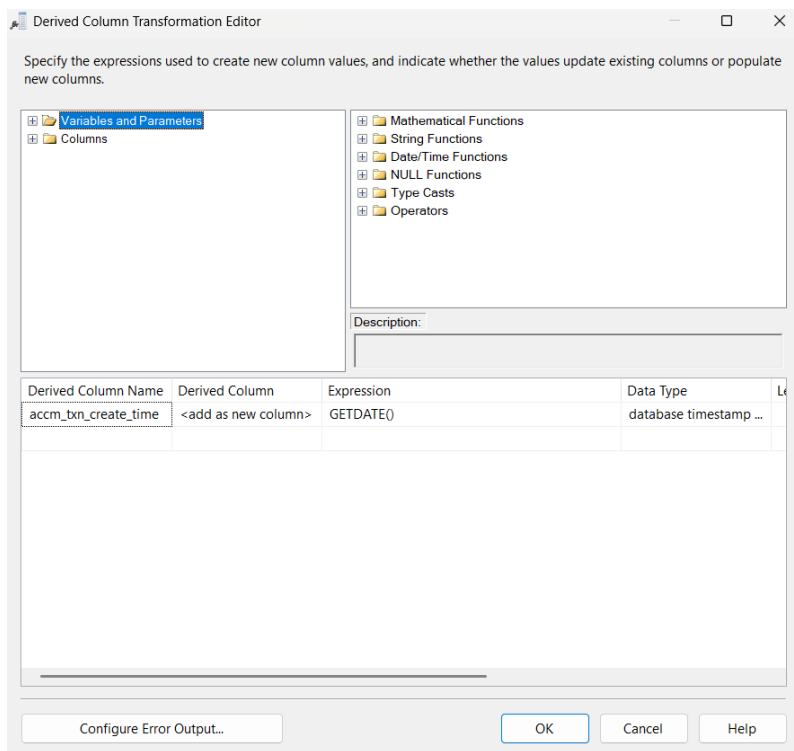
Lookup ShippingSK:



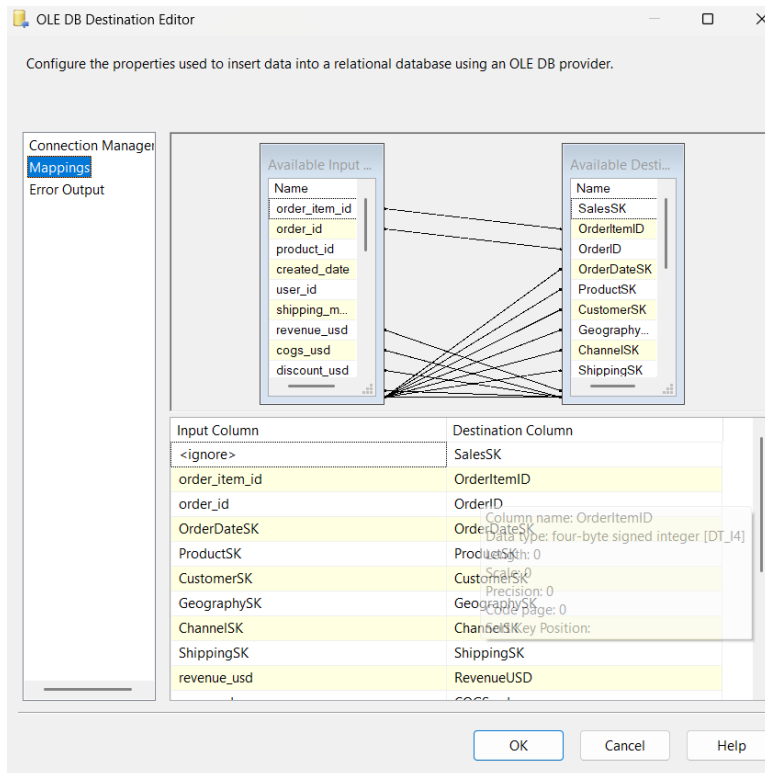
Lookup ChannelSK:



ETL Timestamp:



Load Factsales:



Union All:

Union All Transformation Editor

Configure the properties used to merge multiple inputs into one output by creating mappings between columns.

Output Column Name	Union All Input 1	Union All Input 2	Union All Input 3
order_item_id	order_item_id	order_item_id	order_item_id
order_id	order_id	order_id	order_id
product_id	product_id	product_id	product_id
created_date	created_date	created_date	created_date
user_id	user_id	user_id	user_id
shipping_method	shipping_method	shipping_method	shipping_method
revenue_usd	revenue_usd	revenue_usd	revenue_usd
cogs_usd	cogs_usd	cogs_usd	cogs_usd
discount_usd	discount_usd	discount_usd	discount_usd
refund_amount_usd	refund_amount_usd	refund_amount_usd	refund_amount_usd
is_refunded	is_refunded	is_refunded	is_refunded
country	country	country	country
utm_source	utm_source	utm_source	utm_source
utm_campaign	utm_campaign	utm_campaign	utm_campaign
utm_content	utm_content	utm_content	utm_content
device_type	device_type	device_type	device_type
OrderDateSK	OrderDateSK	OrderDateSK	OrderDateSK

OK Cancel Help

Error Log:

OLE DB Destination Editor

Configure the properties used to insert data into a relational database using an OLE DB provider.

Connection Manager
Mappings
 Error Output

Available Input ...

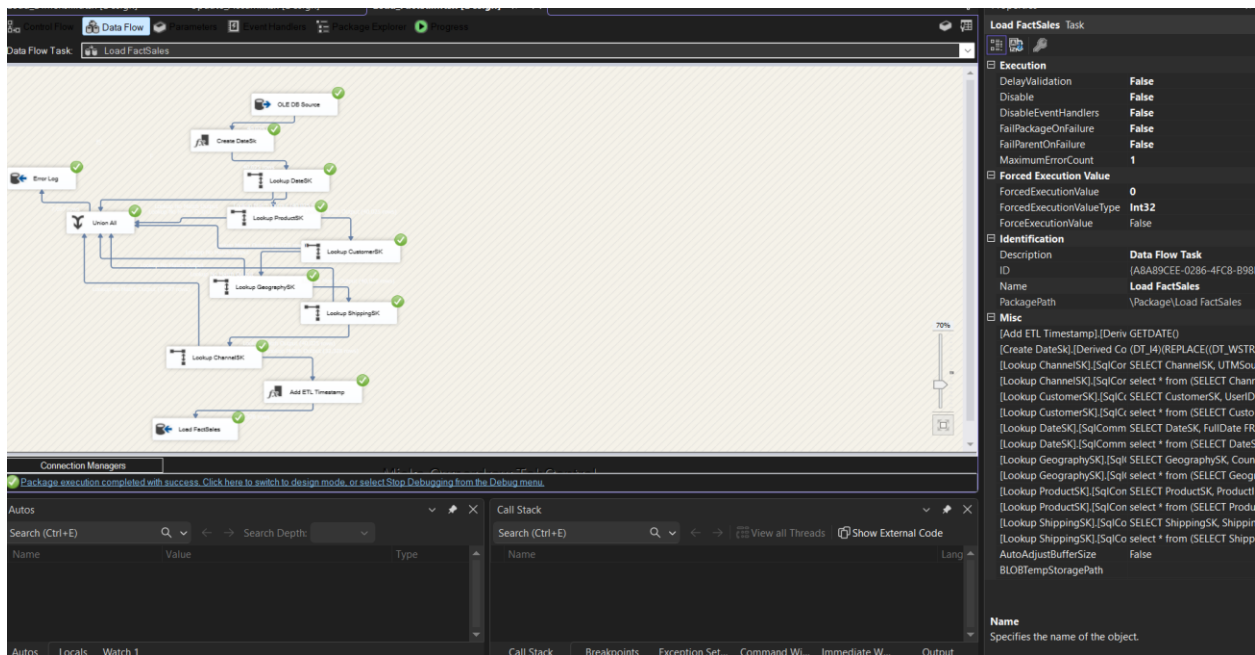
Name
order_item_id
order_id
product_id
created_date
user_id
shipping_m...
revenue_usd
cogs_usd
discount_usd

Available Desti...

Name
OrderItemID
OrderID
error_reason
logged_at

Input Column	Destination Column
order_item_id	OrderItemID
order_id	OrderID
<ignore>	error_reason
<ignore>	logged_at

OK Cancel Help



SQLQuery10...H (sa (80))

```

USE ToyStore_DWH;
-- Count
SELECT COUNT(*) AS FactRows FROM FactSales;
-- expect 40025
-- Error Log
SELECT COUNT(*) AS ErrorRows FROM FactSales_ErrorLog;
-- expect 0
-- Accumulating fact columns check
SELECT TOP 3
  OrderItemID,
  accm_txn_create_time,
  accm_txn_complete_time,
  txn_process_time_hours
FROM FactSales;
-- accm_txn_create_time = filled
-- accm_txn_complete_time = NULL
-- txn_process_time_hours = NULL

```

100 % No issues found Ln: 39, Ch: 52 (954 chars, 39 lines) TABS CRLF UTF-8

Results

OrderItemID	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
94	2026-03-23 02:48:24.390	NULL	NULL
93	2026-03-23 02:48:24.390	NULL	NULL
94	2026-03-23 02:48:24.390	NULL	NULL

SalesSK	FullDate	ProductName	Country	City	UTMSource	ShippingMethod	RevenueUSD	GrossProfit
161655	2012-04-23	The Original Mr. Fuzzy	United Kingdom	New York	gsearch	Standard	49.99	30.50
161665	2012-09-06	The Original Mr. Fuzzy	United States	New York	gsearch	Standard	49.99	30.50
161664	2012-08-04	The Original Mr. Fuzzy	United States	New York	gsearch	Standard	49.99	30.50
161663	2012-09-04	The Original Mr. Fuzzy	Netherlands	Amsterdam	gsearch	Standard	49.99	30.50
161662	2012-08-02	The Original Mr. Fuzzy	United States	New York	gsearch	Standard	49.99	30.50

Query executed successfully. LAPTOP-20K14P2C\MSSQLSERVER... sa (80) ToyStore_DWH 00:00:05 Row: 1, Col: 1 10 rows

Properties

Current connection parameters

Aggregate Status

Connection : 00:00:05.761
Elapsed time : 00:00:05.761
Finish time : 3/23/2026 2:51:09 AM
Name : LAPTOP-20K14P2C\M
Rows returned : 10
Start time : 3/23/2026 2:51:04 AM
State : Open

Connection

Connection : LAPTOP-20K14P2C\M

Connection Details

Connection : 00:00:05.761
Connection : Not encrypted
Connection : 3/23/2026 2:51:09 AM
Connection : 10
Connection : 3/23/2026 2:51:04 AM
Connection : Open
Display name : LAPTOP-20K14P2C\M
Login name : sa
Server name : LAPTOP-20K14P2C\M
Server version : 17.0.1000
Session Trac
SPID : 80
TDS protocol : 0x74000004

Name

The name of the connection.

Task 06: Accumulating Fact Table

An accumulating snapshot fact table tracks a multi-stage business process by updating the same row as each stage completes. Unlike transaction fact tables which insert a new row for every event, accumulating snapshot fact tables update existing rows to record milestone timestamps as the process progresses.

In this implementation, FactSales tracks two stages of the order fulfillment process:

Column	Stage	Description
accm_txn_create_time	Stage 1 — Order Placed	Populated by Package 2 when the order line item is first loaded
accm_txn_complete_time	Stage 2 — Order Completed	Updated by Package 3 when the order completion is recorded
txn_process_time_hours	Computed	Automatically calculated by SQL Server as the difference in hours between the two timestamps

The `txn_process_time_hours` column is a computed column defined as `DATEDIFF(HOUR, accm_txn_create_time, accm_txn_complete_time)`. SQL Server recalculates this automatically whenever `accm_txn_complete_time` is updated — no manual update is required.

Before State:

Before Package 3 ran, FactSales showed the following state for all 40,025 rows:

- `accm_txn_create_time` — populated with the ETL load timestamp
- `accm_txn_complete_time` — NULL (order not yet completed)
- `txn_process_time_hours` — NULL (cannot be calculated until completion time is known)

This is the correct initial state for an accumulating snapshot fact table — the first milestone is recorded and subsequent milestones are updated as they occur.

Step 1: Verify the accm_txn_create_time, accm_txn_complete_time, txn_process_time_hours values are filled, Null, Null.

SQLQuery1.s_H (sa (73))

```
1 USE ToyStore_DWH;
2
3
4 SELECT TOP 3
5     OrderItemID,
6     accm_txn_create_time,
7     accm_txn_complete_time,
8     txn_process_time_hours
9 FROM FactSales;
```

OrderItemID	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
84	2026-03-23 02:48:24.390	NULL	NULL
93	2026-03-23 02:48:24.390	NULL	NULL
94	2026-03-23 02:48:24.390	NULL	NULL

Query executed successfully. LAPTOP-20K14P2C\MSSQLSERVER... sa (73) ToyStore_DWH 00:00:00 Row: 1, Col: 1 3 rows

Step 2: Generate order_completions.csv

This query generates realistic completion times by adding a random number of hours between 24 and 72 to the original create time — simulating real-world order processing times. The results were saved as order_completions.csv at C:\ToyStore\Sources\order_completions.csv.

SQLQuery1.s_H (sa (73))

```
10
11
12
13
14 USE ToyStore_DWH;
15
16 SELECT TOP 100
17     OrderItemID AS order_item_id,
18     DATEADD(HOUR, ABS(CHECKSUM(NEWID()))%48+24,
19     accm_txn_create_time) AS accm_txn_complete_time
20 FROM FactSales
21 ORDER BY SalesK;
```

order_item_id	accm_txn_complete_time
84	2026-03-25 19:48:24.390
93	2026-03-24 04:48:24.390
94	2026-03-24 08:48:24.390
96	2026-03-24 04:48:24.390
95	2026-03-25 14:48:24.390
100	2026-03-24 13:48:24.390
107	2026-03-25 10:48:24.390
114	2026-03-24 23:48:24.390
118	2026-03-25 22:48:24.390
129	2026-03-25 14:48:24.390
137	2026-03-24 23:48:24.390
138	2026-03-25 03:48:24.390
142	2026-03-24 07:48:24.390
147	2026-03-24 20:48:24.390
149	2026-03-25 06:48:24.390
157	2026-03-25 14:48:24.390
162	2026-03-25 23:48:24.390
164	2026-03-24 19:48:24.390
585	2026-03-25 01:48:24.390
687	2026-03-24 06:48:24.390

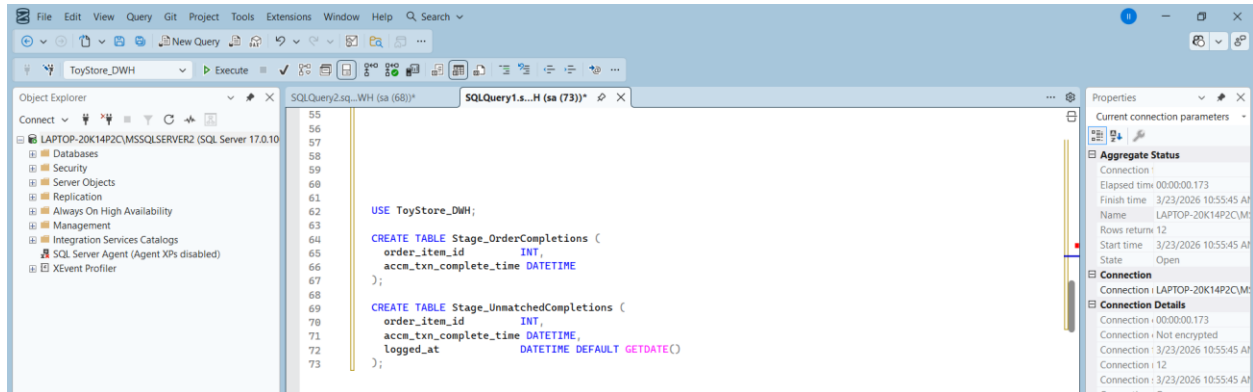
Query executed successfully. LAPTOP-20K14P2C\MSSQLSERVER... sa (73) ToyStore_DWH 00:00:00 Row: 1, Col: 1 100 rows

Step 3: Creating Staging Tables

Two staging tables were created in ToyStore_DWH to support the update process:

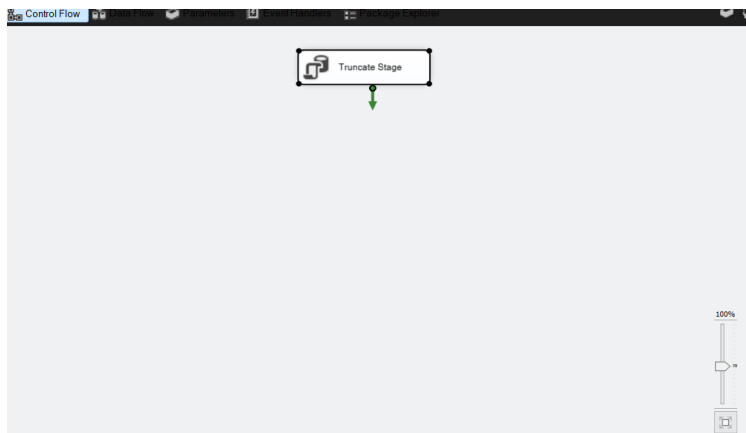
Stage_OrderCompletions — holds the incoming completion records from the CSV file before they are applied to FactSales.

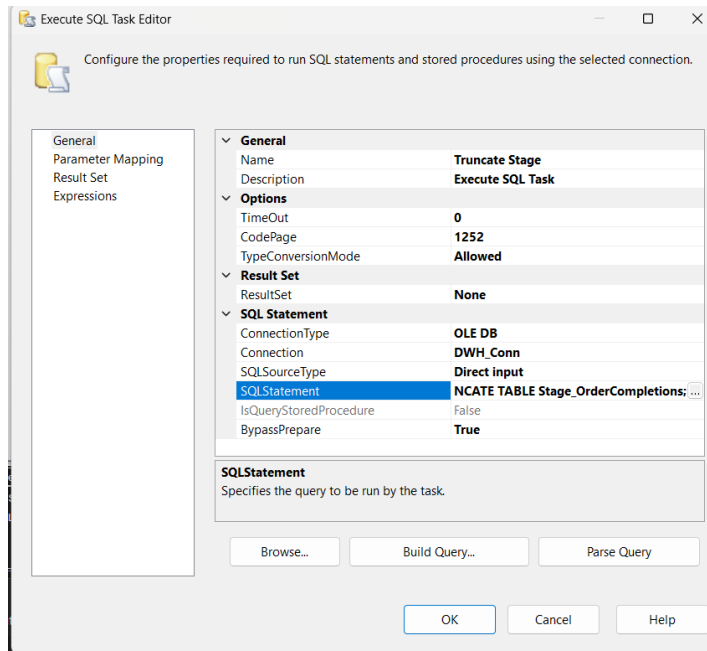
Stage_UnmatchedCompletions — captures any completion records where the order_item_id does not exist in FactSales, providing error handling and audit trail.



Step 4: Build Package 3: Update_AccumFact.dtsx

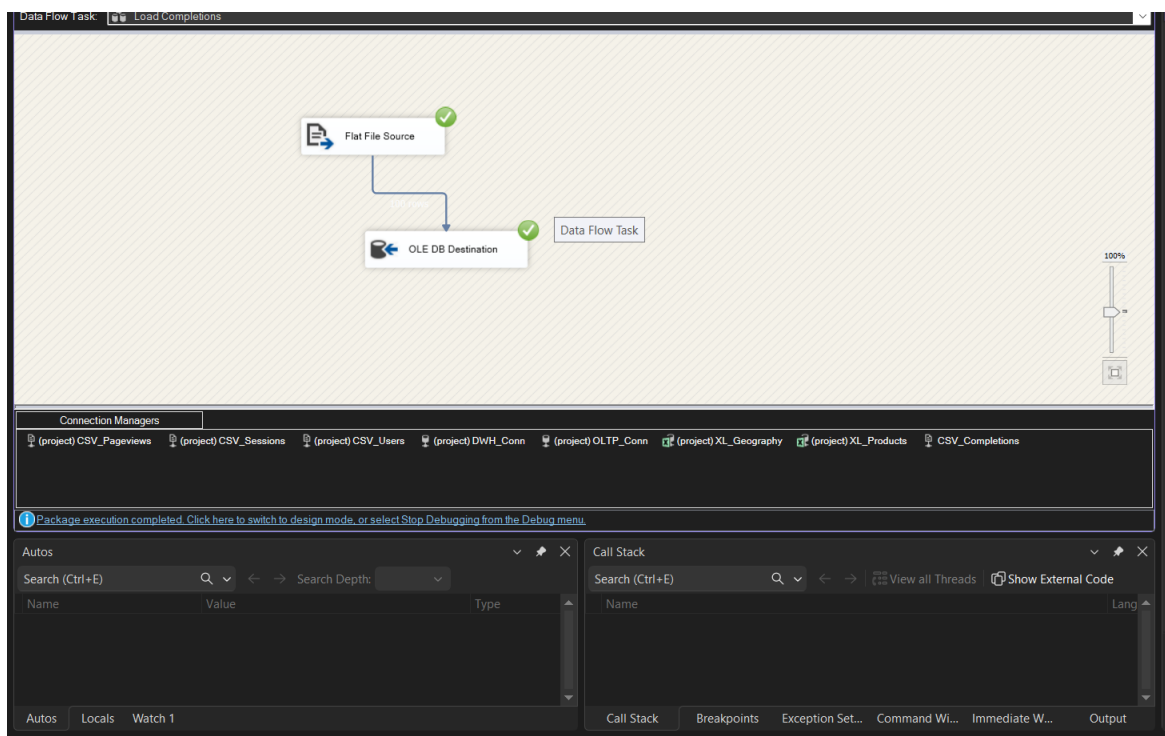
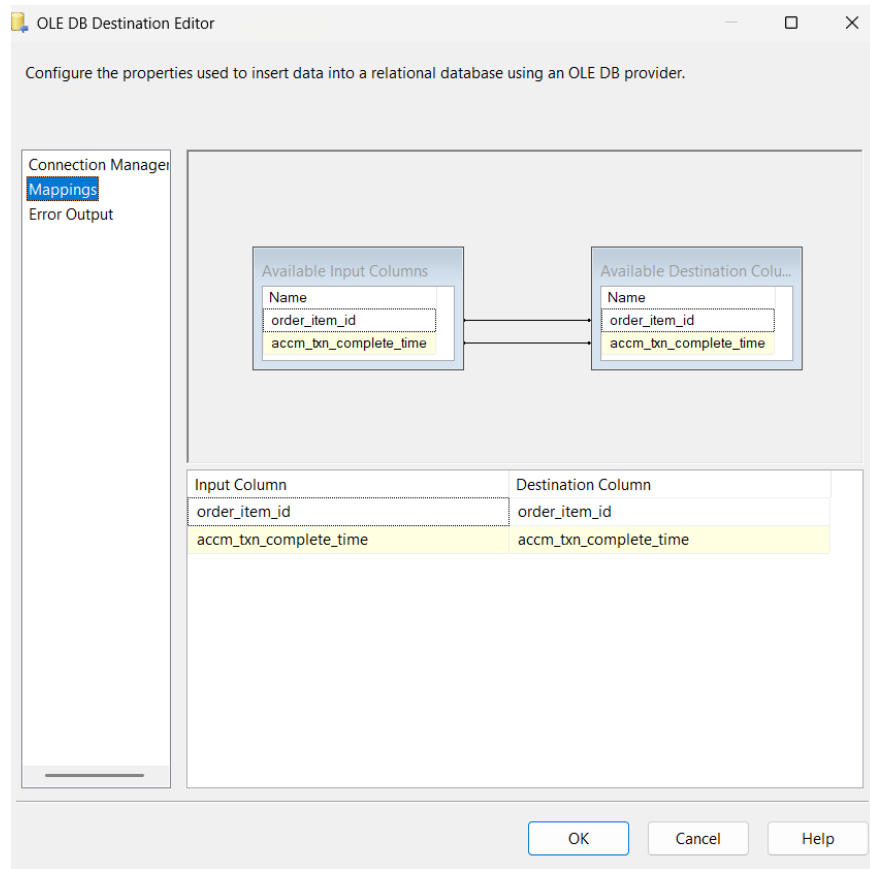
Task 1: Execute SQL Task: Truncate Stage





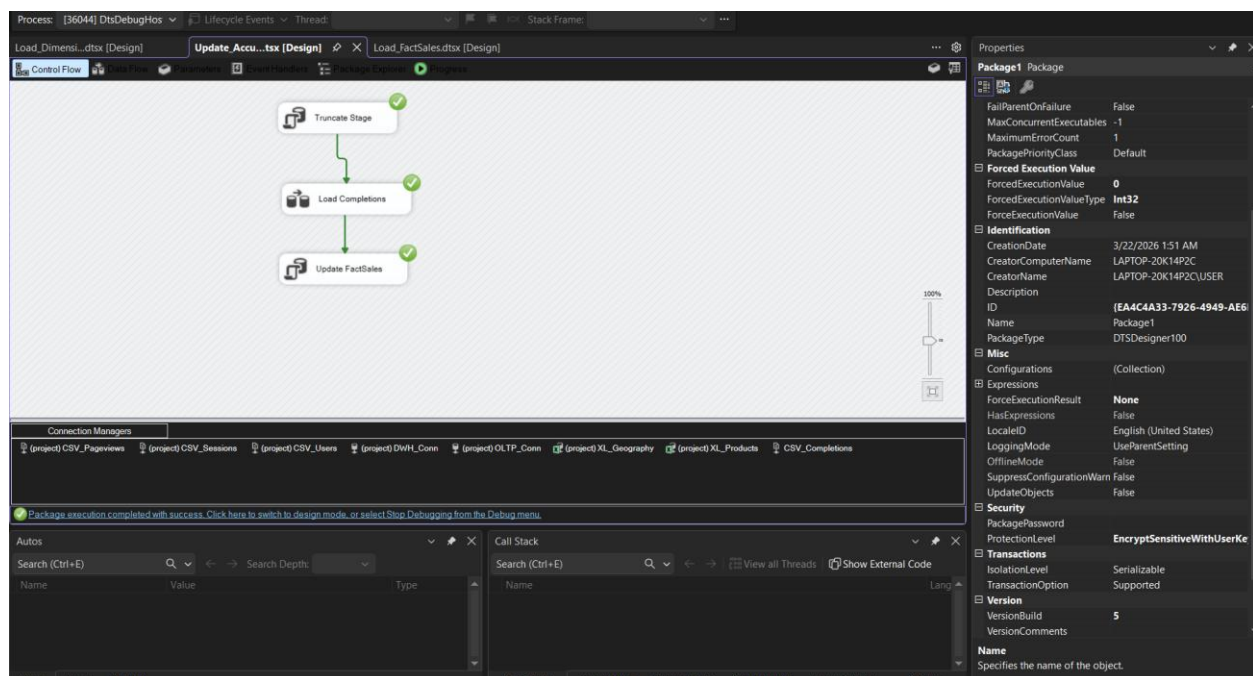
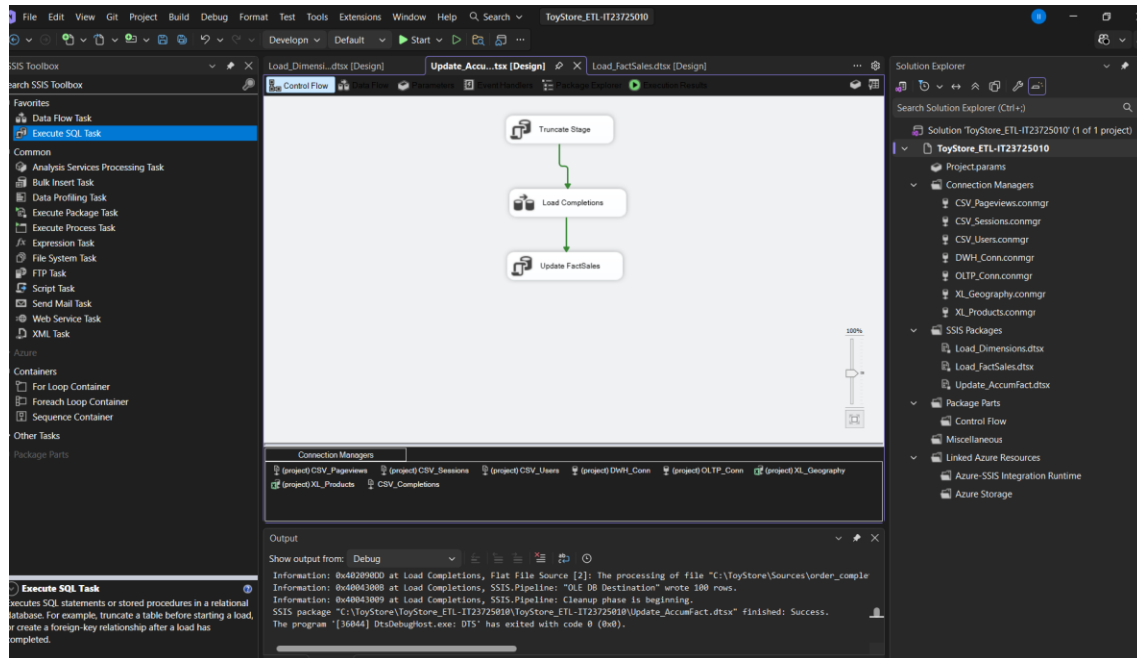
Task 2: Data Flow Task: Load Completions

A Data Flow Task reads the order_completions.csv file using a Flat File Source and loads the records into Stage_OrderCompletions using an OLE DB Destination. The order_item_id column is read as a four-byte signed integer and accm_txn_complete_time is read as a database timestamp.



Task 3: Execute SQL Task: Update FactSales

An Execute SQL Task performs two operations. First it updates FactSales by joining with Stage_OrderCompletions to set accm_txn_complete_time for all matching rows. Second it logs any unmatched records to Stage_UnmatchedCompletions for error tracking.



The screenshot shows the SQL Server Enterprise Manager interface. The central pane displays a query window with the following SQL code:

```

1  USE ToyStore_DWH;
2
3  -- Rows updated (expect 100)
4  SELECT COUNT(*) AS UpdatedRows
5  FROM FactSales
6  WHERE accm_txn_complete_time IS NOT NULL;
7
8  -- Sample check (txn_process_time_hours should be 24-72)
9  SELECT TOP 10
10 OrderItemID,
11 accm_txn_create_time,
12 accm_txn_complete_time,
13 txn_process_time_hours
14 FROM FactSales
15 WHERE accm_txn_complete_time IS NOT NULL;
16
17 -- Unmatched rows (expect 0)
18 SELECT COUNT(*) AS UnmatchedRows
19 FROM Stage_UnmatchedCompletions;

```

The Results pane shows the following data:

OrderItemID	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
84	2026-03-23 02:48:24.390	2026-03-25 19:48:24.390	65
93	2026-03-23 02:48:24.390	2026-03-24 04:48:24.390	26
94	2026-03-23 02:48:24.390	2026-03-24 08:48:24.390	30
96	2026-03-23 02:48:24.390	2026-03-24 04:48:24.390	26
95	2026-03-23 02:48:24.390	2026-03-25 14:48:24.390	60
100	2026-03-23 02:48:24.390	2026-03-24 13:48:24.390	35
107	2026-03-23 02:48:24.390	2026-03-25 10:48:24.390	56
114	2026-03-23 02:48:24.390	2026-03-24 23:48:24.390	45
118	2026-03-23 02:48:24.390	2026-03-25 22:48:24.390	68
129	2026-03-23 02:48:24.390	2026-03-25 14:48:24.390	60

The UnmatchedRows section shows 0 rows.

The Properties pane on the right shows connection details for the current connection.

Summary

The accumulating snapshot fact table pattern was successfully implemented. FactSales now tracks two milestones of the order fulfillment process. The computed column `txn_process_time_hours` automatically calculates the processing duration whenever a completion time is recorded. This design allows business analysts to measure order processing efficiency and identify delays in the fulfillment pipeline.